

Strategic or Scarred? Disparities in College Enrollment and Dropout Response to Macroeconomic Conditions*

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Abstract

Recessions create enduring effects, or scars, on young individuals' careers. I investigate how educational choices amplify or mitigate these scarring effects by income. Low-income young people face dual scarring effects: an increased likelihood of dropping out of college and enduring negative labor market entry effects. High- and middle-income young people strategically evade these repercussions by delaying labor market entry through timely college enrollment during economic downturns. I quantify the lifetime repercussions of experiencing a recession during two critical phases (around high-school graduation and college attendance) using a dynamic life-cycle model with educational choices calibrated to US data. I find that the negative consequences of recessions are largely concentrated on the low-income group.

JEL Codes: *E24, E32, E61, I23, I24, J24.*

Keywords: *Business cycles, unemployment, college, dropout, scarring effects.*

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Macroeconomic conditions are well-known to impact young people’s schooling choices (Gustman & Steinmeier, 1981; Betts & McFarland, 1995; Sakellaris & Spilimbergo, 2000; Dellas & Koubi, 2003; Hérault et al., 2012; Johnson, 2013; Boffy-Ramirez et al., 2013; Smith, 2017; Guo, 2018; Cajner et al., 2021; Blom et al., 2021; Stuart, 2022; Graves & Kuehn, 2022; Schanzenbach et al., 2024; Sadaba et al., 2024; Grenet et al., 2025). However, less is known about how these impacts differ by students’ socioeconomic backgrounds. Labor force entry decisions during recessions involve difficult tradeoffs for recent high school graduates and college students. On the one hand, family financial distress creates a wealth effect that raises the relative cost of staying in college. On the other hand, a weak labor market lowers the opportunity cost of being outside the labor force. This paper focuses on how the relative importance of these mechanisms correlates with family income.

Using CPS micro data, I document the following novel empirical findings. College students from low-income families—those in the bottom 35% of the income distribution—are 10% more likely to drop out during periods of high unemployment. In contrast, students from high-income families—the top 25%—are 4% more likely to persist in college on a month-to-month basis during periods of high unemployment. Furthermore, low-income individuals are 24% less likely to have earned a degree, compared to a 31% higher likelihood among high-income individuals in the long run. Years after graduating high school during times of high unemployment, low-income individuals are only 3% more likely to have attended college, while their high-income peers are 7% more likely. I develop a model with endogenous schooling decisions that mirrors the main tradeoffs that young individuals face during a recession. The model has liquidity constraints and persistent effects of labor market entry conditions on current wages that impact college enrollment and dropout decisions heterogeneously by income levels. I calibrate this model to match key moments for the US using Current Population Survey (CPS) micro data.

The model successfully replicates these empirical findings along two dimensions: the heterogeneous cyclicity of dropout and persistence *while enrolled*, and the relative ordering of counter-cyclical enrollment across income groups *around high school graduation*, with stronger counter-cyclicity for high-income than middle-income. The model’s primary limitation is that it predicts pro-cyclical enrollment for low-income individuals around high school graduation, whereas the data shows near-acyclical enrollment for this group. With the current model ingredients, generating both pro-cyclical dropout while enrolled and counter-cyclical enrollment around high school graduation simultaneously for low-income individuals is not feasible, as the same financial pressure channel operates in both directions. Transparently acknowledging this limitation, I note that constructing a model capable of jointly replicating both patterns — perhaps by incorporating richer heterogeneity in liquidity constraints or allowing parental transfers to respond asymmetrically across the two margins — is technically demanding and remains a worthwhile direction for future research.

Once calibrated to key data moments, the model generates substantial drops in lifetime consumption for the poorest subgroups within the low-income group when a recession occurs during two pivotal periods: around high school graduation and during college attendance. Finally, the model highlights the strategic role of college decisions made by middle and high-income individuals

as a means to mitigate such long-term effects.

I find that low-income individuals are disproportionately scarred by economic recessions during two pivotal time periods. While attending college, an increase in the unemployment rate in a US state equivalent to the rise experienced during the Great Recession of 2009 correlates with an increase in the probability of dropping out of college without earning a degree by 10% on a month-to-month basis. In addition, around high school graduation, the same rise in unemployment rate is correlated with only a 3% rise in the long-run probability of enrolling in college and a 24% reduction in their long-run probability of earning a college degree.

By contrast, high-income individuals strategically evade these scarring effects. When periods of high unemployment coincide with college attendance, they are 4% more likely to complete their college degree, and when such economic downturns coincide with their high school graduation, they are 7% more likely to enroll in college and 31% more likely to graduate.

The substantial increase in college enrollment among high-income individuals after a rise in the unemployment rate in the respondent's state around their high school graduation does not fully translate into a proportionate rise in college degree holders within this income bracket, as the surge in enrollment also brings in individuals who subsequently drop out of college. This spike in college enrollment is followed by an increase in college dropouts among high-income individuals as economic conditions rebound post-recession. I find that this type of college dropout is more prevalent among the male population.

I introduce a dynamic life-cycle individual decision-making model to quantify the long-term effects of experiencing an economic recession during two critical life stages: graduating from high school and entering college. The model also explores how young individuals respond to the tradeoffs posed by these recessions. It incorporates endogenous choices regarding college enrollment, college completion, as well as decisions regarding consumption and savings, all starting from different initial asset levels. Business cycles are generated by a productivity variable subject to a stochastic process. Wages depend both on the current productivity state and the productivity state at the individual's labor market entry, capturing the scarring effects resulting from unfavorable entry conditions. Additionally, the model features liquidity constraints that affect individuals' ability to borrow to pay tuition and living expenses during college enrollment.

The model illustrates that during economic recessions, high-income individuals are more likely to enroll in college or complete their degrees due to lower wages and strong incentives to delay labor market entry. These incentives are further amplified by labor market rigidities, which cause the initial conditions to have persistent effects on future wages. The model explores the following mechanism: for the high-income group, the diminished opportunity cost incentivizes even individuals who might be *a priori* less academically suited for college. Once they realize their unsuitability for college and the labor market has recovered, they might choose to drop out of college. The model successfully replicates this.

I use this same dataset to estimate the following key parameters of the model—the variance of the business cycle, the influence of labor market entry conditions on current wages, and tuition

fees—with key moments of the data. Experiencing a rise in the unemployment rate in a state similar to the rise experienced in the 2009 recession, either while attending college or around the time of high school graduation, significantly reduces the lifetime consumption of individuals from low-income backgrounds, while having a negligible effect on those from high-income groups. I conduct three experiments to assess policy relevance. Among low-income individuals, those with a high probability of college success are most affected by recessions occurring around high school graduation, while among high-income individuals, the least likely to succeed are most impacted since recession incentivizes individuals with high probabilities of dropping out of college to enroll. Additionally, rising tuition costs under tight liquidity constraints and increasing labor market rigidities disproportionately harm low-income groups.

This paper contributes to two branches of the literature.

First, a large empirical literature documents that entering the labor market during a recession generates persistent negative effects on wages and employment. Kahn (2010), Speer (2016) and Schwandt & Von Wachter (2019) show these effects persist for over a decade among US workers, with Kondo (2015) documenting significant heterogeneity across gender and race.¹ These findings are consistent with Beaudry & DiNardo (1991)’s implicit contract model, in which initial entry conditions predict wage trajectories better than current macroeconomic conditions. The theoretical mechanisms behind these persistent entry effects have been explored by Guo (2018) and Acabbi et al. (2022), who emphasize human capital accumulation and job quality matching as key transmission channels. A related strand focusing on job displacement studies Gregg & Tominey (2005), who documents the long-run wage penalties from male youth unemployment, alongside Jarosch (2023) and Huckfeldt (2022), who highlight human capital and job security loss and directed search into lower-skill submarkets as the mechanisms behind the persistence of unemployment scarring. While these displacement studies focus on incumbent workers rather than labor market entrants, they provide conceptual parallels to the mechanisms this paper emphasizes and highlight forces deliberately left outside the model. I contribute to this branch by showing that labor market entry decisions are themselves endogenous: individuals can use college enrollment to strategically postpone entry and avoid scarring effects, and I provide empirical evidence that middle- and high-income individuals do so in economically significant magnitudes.

Second, a separate literature studies the cyclicity of college enrollment. One strand finds that recessions lower the opportunity cost of education and raise enrollment (Gustman & Steinmeier, 1981; Betts & McFarland, 1995; Sakellaris & Spilimbergo, 2000; Guo, 2018; Cajner et al., 2021; Stuart, 2022; Schanzenbach et al., 2024; Sadaba et al., 2024).² A complementary strand shows

¹Similar patterns have been documented internationally: see Bachmann et al. (2010) and Stevens (2008) for Germany, Genda et al. (2010) for Japan, Oreopoulos et al. (2012) for Canada, Brunner & Kuhn (2014) for Austria, Fernández-Kranz & Rodríguez-Planas (2018) and Escalonilla et al. (2021) for Spain, and Van den Berge (2018) for the Netherlands. Oyer (2006) documents persistent entry effects for Ph.D. graduates, and Choi et al. (2020) for college graduates entering during the Asian financial crisis.

²For additional evidence on counter-cyclical college enrollment across different samples and time periods, see Dellas & Koubi (2003), Hérault et al. (2012), Johnson (2013), Boffy-Ramirez et al. (2013), Smith (2017), Blom et al. (2021), Graves & Kuehn (2022), and Grenet et al. (2025).

that negative income shocks can disrupt enrollment for financially constrained students (Coelli, 2011; Ost et al., 2018; Huttunen & Riukula, 2024). Hershbein (2012) documents long-run effects of economic conditions at high school graduation on educational attainment, showing that entering a weak labor market at age 18 has persistent consequences for degree completion. Most closely related to this paper, Anstreicher (2025) uses linked Census-ACS data to study how the Great Recession affected college attainment across the parental income distribution, finding that negative long-run effects are concentrated among middle-income families. His analysis identifies parental home equity losses as the primary mechanism: middle-class homeowners lost their main college-financing resource when housing values collapsed, while poor families — largely renters — were insulated through need-based aid, and wealthy families could self-finance through other assets. This paper complements Anstreicher (2025) in three ways. First, by studying the income flow and liquidity channel across multiple recessions rather than the wealth shock channel specific to the 2007–2009 housing crash, I find that the low-income group bears the largest scarring. Second, I separately identify the enrollment margin from the dropout margin — two forces that approximately cancel for low-income individuals in long-run attainment measures but have sharply opposite income gradients, as Tables 2 and 4 show. Third, I connect these educational responses to the entry scarring literature by documenting that high-income individuals use college enrollment strategically to avoid adverse initial labor market conditions — a channel that attainment-focused studies cannot identify. Together, the two papers suggest that recessions harm different income groups through different channels: the wealth shock channel concentrated on middle-income homeowners, and the income flow and liquidity channel concentrated on low-income individuals.

This paper is structured as follows: in Section 1, I describe the data set and I present preliminary statistics. In Section 2, I show the two main empirical findings of the paper as well as some robustness checks and a discussion of the central identifying assumptions underpinning the empirical section of the paper. In Section 3, I introduce a dynamic individual decision-making model designed to capture the dynamics of college decisions as a response to macroeconomic conditions. I calibrate the model to match key moments in the US data in Section 4 and I present the main results of the model with a comparison to the empirical findings in Section 5. In Section 6, I present the two main counterfactuals of the paper. Finally, Section 7 concludes.

1 Data

I use yearly aggregate data from the US Census to present stylized facts regarding educational choices as a response to macroeconomic conditions. I show that college enrollment has been counter-cyclical since 1970 in the US.

I also use Integrated Public Use Microdata Series (IPUMS) Current Population Survey (CPS) monthly micro data spanning from January 1992 to July 2024, which constitutes the central dataset for the main empirical facts in Section 2.³ I split the data set into two: longitudinal and cross-

³The reason I use the data since 1992 is because the category college dropout was non-existent prior to that date.

sectional data. This dataset is particularly well-suited for examining the tradeoffs faced by young individuals during periods of high unemployment since it encompasses three major economic crises, and crucially, it includes detailed information on college dropouts—an essential variable for understanding educational decisions under economic stress. However, a notable limitation is the relatively short duration for which individuals remain in the sample.

I use the longitudinal data to show the first main empirical finding: low-income individuals are more likely to drop out of college during high unemployment rate periods, while middle- and high-income individuals are more likely to stay in college amid high unemployment rate periods. Further, I observe that middle- and high-income individuals are also more likely to transition from the labor force to college during high unemployment rate periods.

I use the cross-sectional data to show the second empirical finding: high unemployment rates around high school graduation exert varying effects on future college enrollment decisions across different income groups. It leads to an increased likelihood of college enrollment for middle- and high-income individuals, while showing relatively insignificant changes in the probability of college attendance for their low-income counterparts.

1.1 Aggregate Yearly Data

I use historical yearly data from the CPS to examine school enrollment trends in the United States.⁴ The main dependent variable is the annual college enrollment of students under 35 years old relative to the entire student-age population deviations from the trend.⁵ To construct this variable, I aggregate the counts of undergraduates, graduate students, and two-year college students under 35 years old and express it as a percentage of the total student age population in the US. I also apply an HP filter to account for the linear trend of this variable.

I run a linear regression to examine the relationship between the main dependent variable, yearly college enrollment deviations from trend, and key business cycle variables, as specified in equation 1. The data spans yearly observations from 1970 to 2022.

$$y_t = \alpha + \beta_1 \text{ business cycle}_t + \epsilon_t \quad (1)$$

In Table 1, I present the regression results including four independent variables that capture the business cycle phase. These variables include the yearly mean of the unemployment rate, deviations from the NAIRU, a binary indicator denoting the occurrence of a recession and the real GDP yearly growth rate.⁶ To align these business cycle variables with the academic calendar, I converted them so that real GDP in 1990 represents the cumulative sum of real GDP from 1989:III, 1989:IV, 1990:I,

⁴More specifically, I use the Table A.7: College Enrollment of Students 14 Years Old and Over (<https://www.census.gov/data/tables/time-series/demo/school-enrollment/cps-historical-time-series.html>).

⁵I take the following age groups from the United Nations dataset: 15 years old to 19 years old, 20 to 24, 25 to 29 and 30 to 34.

⁶I am using the NBER recession periods. Since the NBER does not use yearly frequency to define recession periods, I define a recession year in period t if there is at least one recession month in the second half of $t - 1$ or the first half of t . For instance, for 2002, if there is a recession month between July of 2001 and June of 2002 both included, 2002 would be considered a recession year. I do this to account for academic calendar decisions.

and 1990:II. This adjustment ensures that the decision to pursue college education commencing September 1990 remains independent of macroeconomic realizations in the subsequent two quarters of that year. As a robustness exercise, I extend the regression analysis to encompass linear and exponential time trends, the results persist unaltered, see Table A1.

Table 1: College enrollment is counter-cyclical

	Effect on college enrollment deviations from trend
Unemployment rate (p.p.)	0.329*** (0.102)
Unemployment rate deviations from NAIRU (p.p.)	0.394*** (0.102)
Recession (binary)	0.243 (0.366)
Real GDP growth (YoY%)	-0.243*** (0.079)

Note: The dependent variable is annual college enrollment of students under 35 as a share of the student-age population, expressed as deviations from an HP-filtered trend. Business cycle variables are adjusted to align with the academic calendar so that year t reflects cumulative conditions ending in the second quarter of t . The sample standard deviation of the HP-filtered enrollment series is 0.956 percentage points. Source: CPS, World Bank and UN population data, Federal Reserve Bank of St. Louis. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

College enrollment is strongly counter-cyclical. This effect is both statistically and economically significant. An average national unemployment rate increase akin to the magnitude experienced during the last Great Recession, approximately 5.6 percentage points, would be associated with an average increase of 1.9 standard deviations of yearly college enrollment deviations from the long-run trend.

In conclusion, college enrollment seems counter-cyclical on aggregate. The underlying intuition suggests that during high unemployment rate periods, labor market outcomes deteriorate, thereby reducing the opportunity cost of pursuing higher education. However, high unemployment rate periods may simultaneously exert negative effects on individuals' liquidity, potentially hindering their ability to afford substantial tuition costs. Section 2 is dedicated to probing these two channels for different income levels.

1.2 Micro CPS Data

I use IPUMS CPS monthly data spanning from January 1992 to July 2024. The CPS adopts a rotating panel design, wherein each individual appears for consecutive 4-month periods, followed by an 8-month absence from the sample, before potentially reappearing for another 4 months. I only keep observations for individuals aged at least 16 years old.

I use two data sets: one for longitudinal analyses, examining individuals' transitions both into and out of college, as well as the transitions from the labor force to college, and the other dedicated

to cross-sectional analysis. I present descriptive statistics for key variables in Table A2, where I applied the weights using the variable *wtfinl* to account for the sample’s representativeness.⁷

For the longitudinal data, the first two columns of the table, I keep an average of nearly four observations per individual to facilitate the study of pertinent transitions. I drop observations for individuals appearing after the 8-month absence in order to better identify the relevant transitions. Each individual is assigned the income bracket observed at their first appearance in the sample, which is then held fixed throughout their subsequent monthly observations.⁸ In the cross-sectional data set, in the last two columns, I adopt a cross-sectional approach and keep only one observation per individual. Furthermore, to discern the heterogeneous effects across different family income levels, I limit the second data set to a sub-sample comprising individuals aged 25 or younger.⁹

I only include observations from individuals who have responded to the questionnaire called *SCHLCOLL* in order to distinguish people who are currently enrolled in college or not. This variable only included people from ages 16 to 24 until 2012 and from ages 16 to 54 from 2013 onward.

The variable “family income” captures the aggregate income received by all members of the respondent’s family over the past 12 months. I focus all my analysis on young individuals to better proxy this “family income” variable as external for the young individual and to avoid possible endogeneity issues regarding the relationship between own earnings and educational choices. The questionnaire explicitly defines the components comprising this income, including money derived from employment, net business or rental income, pensions, dividends, interest, social security payments, and any other monetary inflows received by family members aged 15 years or older. Given its categorical nature and the change in criterion as of October 2003, I categorize this variable into three distinct groups with comparable weights. The sample divides into three income cohorts as follows: low-income individuals, constituting slightly more than a third of the young sample, comprise individuals with an annual family income below \$30,000. Middle-income individuals, representing also slightly more than a third of the young sample, encompass those with family incomes ranging between \$30,000 and \$75,000. Lastly, high-income individuals, accounting for around a fourth of the young sample, pertain to those with an annual family income exceeding \$75,000.¹⁰

The variable “education” (*EDUC*), extracted from the CPS dataset, provides the educational attainment levels prevalent within the sample. I only keep the period from 1992 because there was a change in the categorical groups included in January 1992. In Table A3, I present a comparative analysis of this variable between the two distinct sample periods, including the option “Some college

⁷All econometric analysis presented in the paper will be using these weights as suggested by IPUMS CPS.

⁸This avoids any concern that income group classification might itself be affected by the enrollment or dropout transitions being studied. I also check that switches from an income bracket to another are infrequent. There are only 6.4% of individuals, out of those who appear 4 times, that change income category within this time range.

⁹Family income variable also includes own individual’s earnings. However, at young ages this correlation is close to 0.

¹⁰These brackets of income are in current \$ terms, so they are not normalized because I do not observe the exact amount, but just the bracket in which an individual belongs to. This should not be an issue given that the brackets are sufficiently broad. I also control for the year of the interview and for time trends within the income group in the main regressions shown in Section 2.

but no degree” that is only available from January 1992 onward.

In Figure A.2, I illustrate the evolving trends in the three principal educational level groups over time. The percentage of individuals with a high school diploma or less accounted for approximately 68% of the population under 25 years old in 1992. However, this figure has subsequently decreased and now stands at 57%. Conversely, the percentage of individuals with a bachelor’s degree or higher constituted a mere 10% of the under 25 years old population in 1992, but this proportion has risen significantly to almost 20% in the last decade. The intermediate group includes individuals with some higher education but without a bachelor’s degree. This education category has remained quite stable around 23%.

2 Main Empirical Findings

I present the central empirical findings of the paper in this section regarding the differential responses of college enrollment and completion to the changes in the unemployment rate, contingent on family income groups. High unemployment rates exert varying impacts on individuals’ decisions concerning college enrollment based on their family income. These empirical findings are key to better understand the tradeoffs that young individuals face during a recession. Particularly, the negative wealth effect, operating through liquidity constraints, and the strategic efforts to avoid scarring effects stemming from bad labor market entry become pronounced during high unemployment rate periods. The negative wealth effect becomes notable as it compels individuals with lower incomes to prematurely drop out from college. Conversely, high-income individuals are more likely to enroll in college, seeking to avoid the scarring effects associated with bad labor market entries. In Section 2.1, I analyze responses following a rise in the unemployment rate in a state while being enrolled in college or in the labor force, whereas in Section 2.2, I analyze responses following a rise in the unemployment rate in a state around high school graduation. Notably, in all analyses including different income groups, the dataset is limited to individuals aged below 25 years, mitigating the potential correlation between personal income and overall family income at older ages.

2.1 Low-Income Are Scarred and High-Income Are Strategic

Before analyzing the heterogeneous responses of experiencing high unemployment rates while being enrolled in college for young individuals, it is important to quantify the magnitude of the scarring effects of experiencing a bad labor market entry. I estimate the following regression model which brings similar results as Schwandt & Von Wachter (2019):

$$y_{i,s,t} = \beta_0 + \beta_1 u_{i,s}^{HS} + \Gamma \mathbf{X}_{i,s,t} + \epsilon_{i,s,t}, \quad (2)$$

where $y_{i,s,t}$ is the outcome variable: log of real earnings in 1992 terms for individual i , in state s in year t or a binary equals 1 if the individual was unemployed t years after own high school graduation, $u_{i,s}^{HS}$ is the unemployment rate in a state around high school graduation date and $\mathbf{X}_{i,s,t}$

are controls such as sex, race and the year of the interview.¹¹

Figure 1 illustrates the dynamic evolution of the coefficient β_1 over t periods following high school graduation for individuals who did not pursue higher education and were employed at the time of the interview. A 1 percentage point (p.p.) increase in the unemployment rate in a state around high school graduation is associated with a significant decline in average real earnings for the employed population persisting for over a decade after high school completion. There is also an approximate 0.5 p.p. immediate increase in the probability of experiencing unemployment that vanishes relatively more quickly.

I repeat the same analysis for individuals who graduated from college in Figure A.3. I restrict the sample to individuals who have a college degree but not more in order to better approximate their college graduation date, 4 years after their 18th birthday. The effects of the unemployment rate in the respondent's state on real earnings are markedly mitigated. The results are consistent with intuitive expectations, where a high school graduation coinciding with a period of high unemployment rate leads to adverse impacts on earnings and unemployment probabilities. In contrast, enrolling in college during economic downturns presents the advantage of postponing labor market entry, thereby evading these deleterious and enduring consequences. Further, the negative labor market entry effects seem to virtually vanish for the college sample.

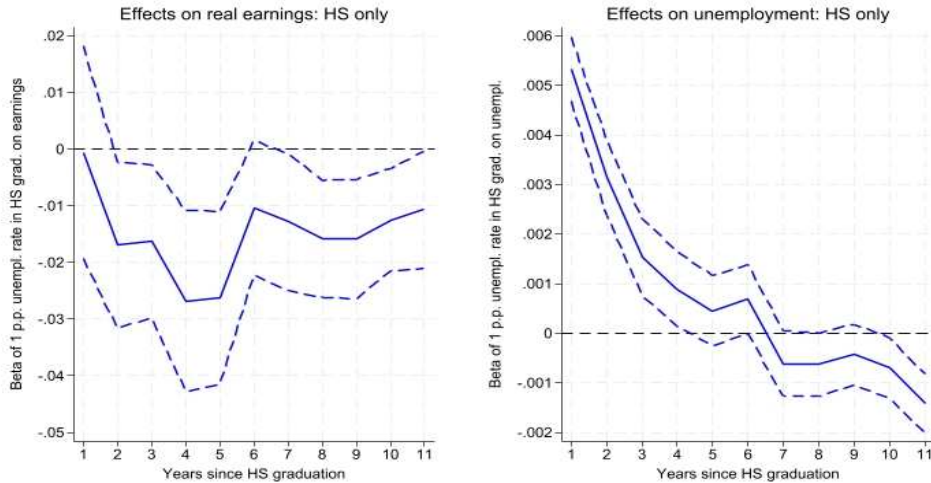


Figure 1: Negative and persistent effects of graduating during recessions.

Note: Each panel plots $\hat{\beta}_1$ from separate regressions by years since high school graduation, estimated on individuals with only a high school diploma, so that high school graduation closely approximates labor market entry. The left panel outcome is log real earnings (1992 dollars) conditional on employment; the right panel outcome is a binary for unemployment. Dashed lines are 95% confidence intervals using robust standard errors.

These findings are economically significant. For instance, the accumulated earnings losses from entering the labor force during a recession similar to the Great Recession would result in a substan-

¹¹I use the average unemployment rate in a state from January to May of the graduation year as a proxy for $u_{i,s}^{HS}$.

tial 8.7% decline in real earnings over the first decade of employment, even when accounting for equivalent employment probabilities. This value is computed by summing the first 10 coefficients and multiplying by 5.6, reflecting the 5.6 percentage point increase in the national unemployment rate during the 2009 recession. Additionally, entering the labor force during an economic state akin to the 2009 Great recession would be linked to a 0.54 p.p. higher likelihood of experiencing unemployment during the ten years following high school graduation. Thus, accounting for the higher probability of being unemployed, entering the labor market during a recession similar to the 2009 one is correlated with a significant reduction in real earnings of around 9.14% for the first decade after entry.¹² These detrimental effects of entry are comparatively lower and less persistent for college graduates, as depicted in Figure A.3.

Based on these findings, I examine whether young individuals from richer families tend to enter the labor force during better macroeconomic conditions when compared to those from relatively disadvantaged households. I use the longitudinal data shown in Section 1.2. I observe an individual for up to four consecutive months, then she disappears for the subsequent 8 months and appears again in the sample for four more consecutive months. I only keep the first up to four appearances for all individuals; nevertheless, the results are robust to maintaining the full sample. The subsequent panel regressions are conducted as follows:

$$z_{i,s,t} = \alpha_0 + \alpha_1 u_{s,t} + \alpha_2 \text{Mid. Inc.} \times u_{s,t} + \alpha_3 \text{High. Inc.} \times u_{s,t} + \Gamma \mathbf{X}_{i,s,t} + \nu_{i,s,t}, \quad (3)$$

where $z_{i,s,t}$ is a binary that takes value 1 if the individual has transitioned from being enrolled in college to not be currently enrolled in that particular month t , and 0 if she stayed in school. That is, an individual appearing four consecutive months, if she was working in the first period, then enrolled in college in the second period and she kept enrolled during the third month and finally dropped out of college in the last month, $z_{i,s,t}$ would be: $z_{i,s,t} = N/A$ for $t = 1$, $z_{i,s,t} = 0$ for $t = 2$, $z_{i,s,t} = 0$ for $t = 3$ and $z_{i,s,t} = 1$ for $t = 4$. If she is not enrolled nor has she been enrolled during this time range, then $z_{i,s,t}$ is missing. Another example, if an individual appears to be enrolled in college the first month and then she drops out in the second period we would have: $z_{i,s,t} = 0$ for $t = 1$, $z_{i,s,t} = 1$ for $t = 2$ and $z_{i,s,t} = N/A$ for $t = 3$ and $t = 4$. I also run the regression for which $z_{i,s,t}$ takes value of 1 if the individual has transitioned from the labor force to college and 0 if she has stayed in the labor force.

$u_{s,t}$ is the current unemployment rate in the respondent's state s during the interview.¹³ I control for sex, race, age, income group, binary of the state, a coefficient capturing a linear time trend and another for the interaction of this time trend with each income group.¹⁴ From this analysis I exclude people who complete the degree, therefore only considering people who leave

¹²I calculate this value by multiplying the real earnings losses conditional on being employed by the probability of being employed (0.9946). I assume that with the remaining probability earnings are 0.

¹³I also run the same regressions using the lag of the unemployment rate as well. The results are robust to these specifications as shown in Table A5.

¹⁴I also use a set of binaries for years and these binaries interacted with each income group instead of imposing a linear trend and the results are very similar.

college without earning the degree.

Table 2: Transitions from enrolled to dropout and from labor force to enrolled

	College Dropouts		LF → College	
	(1)	(2)	(3)	(4)
$u_{s,t}$	0.186** (0.082)	0.147** (0.063)	-0.310** (0.156)	-0.093 (0.080)
Mid. Inc. $\times u_{s,t}$	-0.231*** (0.073)	-0.196*** (0.062)	0.663*** (0.131)	0.525*** (0.104)
High. Inc. $\times u_{s,t}$	-0.244*** (0.083)	-0.208*** (0.071)	0.910*** (0.182)	0.665*** (0.139)
Obs.	706,702	706,702	2,024,651	2,024,651
R-squared	0.000	0.004	0.006	0.015
Controls	No	Yes	No	Yes
Test Middle ($\alpha_1 + \alpha_2$)	-0.045	-0.049	0.353***	0.431***
Test High ($\alpha_1 + \alpha_3$)	-0.058*	-0.061**	0.600***	0.572***

Note: Panel regression using random effects. In columns (1)–(2), the dependent variable equals 1 if the individual transitions from enrolled to not enrolled in a given month, and 0 if she remains enrolled; individuals who complete their degree are excluded. In columns (3)–(4), the dependent variable equals 1 if the individual transitions from the labor force into college enrollment. *Test Middle* and *Test High* report $\hat{\alpha}_1 + \hat{\alpha}_2$ and $\hat{\alpha}_1 + \hat{\alpha}_3$, the total effect for middle- and high-income individuals. Effects in percentage points relative to staying in college or staying in the labor force. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

I present the results of the panel regression in Table 2 and the results of the pool cross-section regression in Table A4 which are almost identical. In columns (1) and (2), I show that individuals from lower income backgrounds are more susceptible to experiencing scarring effects generated by periods of high unemployment rate periods. Specifically, the positive coefficient $\hat{\alpha}_1 > 0$ indicates that, on average, low-income students are more prone to drop out of college during periods of high unemployment rates. Conversely, the coefficients $\hat{\alpha}_1 + \hat{\alpha}_2 < 0$ and $\hat{\alpha}_1 + \hat{\alpha}_3 < 0$ reveal that middle-income and high-income students actually are more likely to stay enrolled in college during high unemployment rate periods.

The results in columns (3) and (4) indicate that low-income individuals exhibit a negative or no significant change in their likelihood of transitioning from the labor force to college during unfavorable economic conditions, whereas their middle and high-income counterparts are substantially more likely to undertake such transitions.

These findings are also economically significant. A rise in the unemployment rate equivalent to the 2009 recession is associated with a 0.8 percentage point increase in the dropout probability for low-income individuals (0.147*5.6), constituting 10% of their mean, while middle- and high-income individuals are 0.3 percentage points less likely to drop out, 3–4% of their respective means. For labor force to college transitions, the same shock produces virtually no change for low-income individuals (−0.5 pp, −3% of mean), while middle- and high-income individuals become 2.4 and 3.2 percentage points more likely to transition into college (15% and 13% of their respective means).

In Table A6 I show the analysis decomposed by sex and we see virtually no differences between

male and female respondents. Further, Tables A7 and A8 present a parallel analysis employing the max and the mean unemployment rate in a state including the 4 months leading up to the census interview leading to indistinguishable results.

These empirical findings show the importance of the negative wealth effect impacting mainly low-income individuals by compelling them to prematurely drop out of college. This often forces them into the labor force during unfavorable economic conditions, thereby incurring the scarring effects shown in Figure 1. Intuitively, this wealth effect appears less present for high-income enrollees, as they appear more likely to persist in college even amidst periods of high unemployment rates. Moreover, high-income individuals exhibit strategic behavior within the labor force, choosing to enroll in college during economic downturns, strategically optimizing their entry conditions. In summary, low-income individuals bear the lasting scars of economic recessions—stemming from adverse labor market entries and the inability to complete their degrees due to financial distress—while high-income individuals strategically avoid these scars by leveraging college enrollment.

2.2 College Enrollment and Completion Disparities

In general, high-income individuals are more likely to go to college. In Table 3, I provide an insight into the educational composition disparities across different income groups. In the upper section of the table I show college disparities and focus on these three groups: high-school diploma only, gone to college and possess a college degree, that last one being a subset of the gone to college group. In the lower section of the table I exclude currently enrolled students to address the issue of including current students within the subcategory of “some college but no degree”, who are yet to complete their degrees. I focus on two groups: high-school diploma only (but not currently enrolled in college) and college dropouts.¹⁵

For each of these sections I show the proportions of each group for low-income, middle-income and high-income and at the bottom of each section I compute disparity ratios. Specifically, the proportion of young individuals with college attendance relative to those with only a high school diploma is 1.94 times higher for high-income individuals than for their low-income counterparts. In comparison, this difference is relatively smaller at 1.19 times higher for middle-income individuals compared to low-income individuals.

The disparities in educational attainment have significant implications for lifetime earnings, with high-income young adults generally exhibiting a higher likelihood of possessing college degrees or some college education, leading to comparatively higher average earnings. Figure 2 provides a detailed illustration of the average real earnings trajectory by age for each educational group. The chart starts at age 16 for the “high school only” group, age 20 for college dropouts, and age 22 for college degree holders. Notably, all educational groups reach their earnings peak around the

¹⁵A potential concern is that late high school graduation — more common among low-income individuals — may inflate the high school diploma share at younger ages and distort disparity ratios. Table A9 replicates Table 3 restricting the sample to individuals aged 21–25, by which age late graduation is essentially complete. The disparity ratios are qualitatively unchanged and if anything larger, confirming that the results are not driven by endogenous timing of high school graduation.

Table 3: Composition of education by income groups

Including enrolled students			
% within income group	HS diploma only	Gone to college	College degree
low-income	29.16	33.96	6.77
middle-income	26.09	36.28	9.10
high-income	18.75	42.28	12.23
Disparity Ratios		College / HS	Degree / HS
Mid. inc. / low-income		1.19	1.50
High inc. / low-income		1.94	2.81
Excluding enrolled students			
% within income group	HS diploma only	Dropout	
low-income	32.66	11.96	
middle-income	28.27	11.75	
high-income	19.76	10.73	
Disparity Ratios		Drop. / HS	
Mid. inc. / low-income		1.15	
High inc. / low-income		1.48	

Note: Sample restricted to individuals aged 18–25. Row shares within each income group do not sum to 100% because the remaining fraction consists of individuals who have not yet completed high school. The bottom panel excludes currently enrolled students to avoid classifying active students as dropouts. Disparity ratios compare the college-to-HS ratio for middle- or high-income individuals relative to the same ratio for low-income. All statistics use CPS sampling weights. Source: IPUMS CPS.

ages of 45 to 50. It is intuitive that college dropouts initially earn less than individuals with no higher education due to their lack of experience; however, they eventually surpass the earnings of their non-college-educated counterparts by 4% over their lifetime. Meanwhile, college degree holders earn, on average, a remarkable 62% more than those without a college education during their lifetime. In Figure A.4 I also provide an extensive examination of the occupational sorting patterns among young adults with different educational backgrounds across 79 distinct occupational categories within the dataset (all except the military, which I excluded). I show that individuals with college education tend to sort into better paid and less volatile occupations on average.

Considering the substantial variations observed in lifetime earnings and occupational choices across distinct educational groups, although I am not suggesting a direct causal relationship between obtaining a college degree and real earnings from the previous charts, I explore the potential effects of periods of high unemployment rates on subsequent college enrollment for individuals belonging to different income strata. I analyze the micro IPUMS CPS monthly dataset spanning from January 1992 to July 2024. Employing cross-sectional data, I estimate the following linear probability model regression that shows the echo effect of having experienced high unemployment rates during your high school graduation on your future educational attainment. Note that unlike in Section 2.1, where the dependent variable showed month-to-month transitions, here it does not carry a

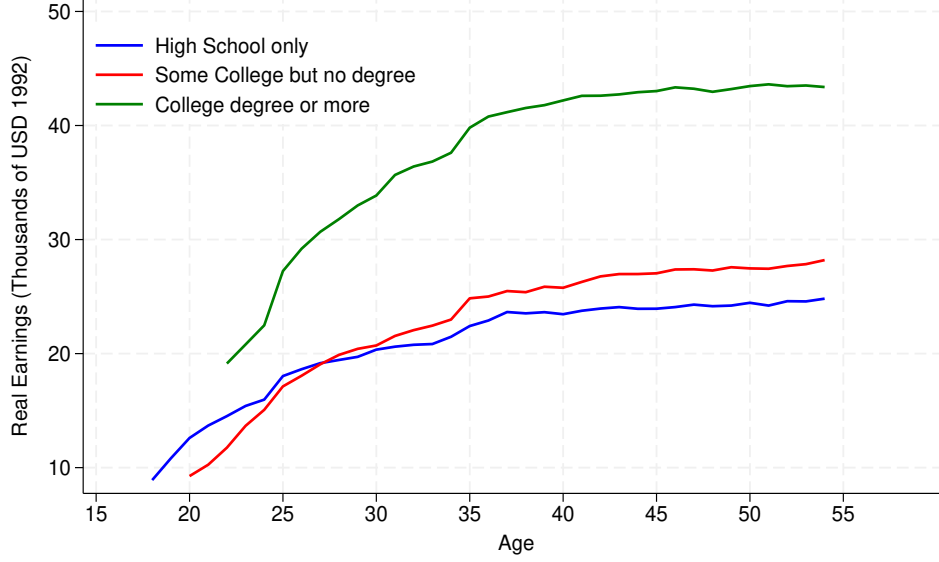


Figure 2: Lifetime real earnings by educational group

Note: Mean real earnings in thousands of 1992 US dollars by single year of age. Currently enrolled students are excluded to avoid misclassifying active students as dropouts. The high school only series starts at age 16, the college dropout series at age 20, and the college degree series at age 22, reflecting typical labor market entry ages for each group. The figure is descriptive and does not imply a causal effect of education on earnings. All series use CPS sampling weights. Source: IPUMS CPS, January 1992 to July 2024.

dynamic interpretation.

$$y_{i,s,t} = \beta_0 + \beta_1 u_{i,s}^{HS} + \beta_2 \text{Mid. Inc.}_{i,s,t} \times u_{i,s}^{HS} + \beta_3 \text{High. Inc.}_{i,s,t} \times u_{i,s}^{HS} + \Gamma \mathbf{X}_{i,s,t} + \epsilon_{i,s,t}, \quad (4)$$

where $y_{i,s,t}$ is a binary that equals 1 if an individual i from state s in period t has gone to college at some point in their lifetime relative to having only a high school diploma, and 0 otherwise. I also explore in separate regressions the probabilities of being a college dropout and a college graduate. The unemployment rate in the respondent's state around high school graduation, denoted as $u_{i,s}^{HS}$, is considered as the key variable of interest, and it is interacted with the family income group. $\mathbf{X}_{i,s,t}$ are control variables such as age, race, sex, a linear time trend at the yearly level and family income group. Specifically, I focus on estimating the coefficients β_2 and β_3 , which represent the interaction effects of the unemployment rate in the respondent's state around high school graduation with the family income group. For the low-income group, the coefficient β_1 represents the marginal effect of the unemployment rate in the respondent's state around high school graduation, while for the middle-income group, it is $\beta_1 + \beta_2$, and for the high-income group, it is $\beta_1 + \beta_3$. It is noteworthy that the subscript t is absent as this analysis is cross-sectional, examining variations across individuals.

Since the dataset does not provide precise graduation dates, I construct a proxy for them. I

rely on the year of the interview and the individual's age, assuming that high school graduation occurs at 18 years old. Accordingly, I use the average unemployment rate in the respondent's state during January to May of the graduation year as a proxy for the macroeconomic conditions at that time.¹⁶ Furthermore, I adopt the assumption that an individual's family income group at the time of the interview reflects their income group at the time of high school graduation. This assumption should not be problematic given the age range of individuals considered, between 18 and 25 years old, and the relatively broad nature of family income categories, which do not undergo substantial changes within a few years.

Table 4: College enrollment and completion and the business cycle

	Attended college (1)		College Degree or more (2)		College Dropout (3)	
$u_{i,s}^{HS}$	0.169*** (0.060)	0.191 (0.121)	-0.002 (0.063)	-0.286** (0.112)	0.226*** (0.073)	0.260** (0.108)
Mid. Inc. $\times u_{i,s}^{HS}$	0.547*** (0.083)	0.356 (0.234)	0.728*** (0.094)	0.284 (0.223)	0.366*** (0.105)	0.281* (0.223)
High. Inc. $\times u_{i,s}^{HS}$	0.850*** (0.083)	0.363 (0.255)	2.065*** (0.114)	0.969*** (0.192)	0.777*** (0.132)	0.438* (0.221)
State Controls	No	Yes	No	Yes	No	Yes
Clustered SE state	No	Yes	No	Yes	No	Yes
Obs.	665,552	665,552	364,787	364,787	324,353	324,353
R-squared	0.020	0.059	0.045	0.263	0.006	0.029
Test Middle ($\beta_1 + \beta_2$)	0.716***	0.547***	0.726***	-0.003	0.592***	0.541***
Test High ($\beta_1 + \beta_3$)	1.019***	0.554***	2.063***	0.683***	1.004***	0.698***

Note: Cross-sectional linear probability model. The dependent variable equals 1 if the individual has ever attended college (column 1), holds a college degree (column 2), or attended college without completing a degree (column 3), each relative to having only a high school diploma; the different observation counts reflect these distinct comparison samples. $u_{i,s}^{HS}$ is the average state unemployment rate from January to May of the year the individual turned 18, used as a proxy for conditions at high school graduation. *Test Middle* and *Test High* report $\hat{\beta}_1 + \hat{\beta}_2$ and $\hat{\beta}_1 + \hat{\beta}_3$. Even-numbered columns include state fixed effects; odd-numbered do not. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level in even-numbered columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results in Table 4 document a striking divergence across income groups. A rise in the unemployment rate equivalent to the 2009 recession is associated with only a 3.1% relative increase in college attendance for low-income individuals and a 23.6% decrease in their probability of earning a college degree. For high-income individuals, the same shock is associated with a 7.4% increase in college attendance and a 31.3% increase in degree completion. Middle-income individuals fall between these extremes, with an 8.4% increase in attendance and negligible change in degree completion. All groups are more likely to have become college dropouts following a high unemployment

¹⁶The analysis is replicated with alternative specifications, utilizing both the maximum unemployment rate between January and May, and solely the unemployment rate in May. Robustness checks show that the results hold under each of these specifications. This choice of months is deliberate, as it coincides with the period when recent graduates initiate their job search or submit college applications. Examining macroeconomic conditions later in the year would not capture these crucial decision-making processes as effectively.

shock around graduation, but the effect is substantially larger for high-income individuals, reflecting the marginal enrollees who enter college during downturns without completing it.

Using the ratios presented in Table 3 I present a numerical exercise using the linear probability model coefficients presented in Table 4. In Table A10 I show the baseline disparity ratios across: college enrollment vs. high school only, college dropout vs. high school only, and college degree holder vs. high school only. Within each disparity group, I compare the ratios between middle and low-income, as well as between high and low-income. The first column presents the average disparity ratio, while the second column represents the disparity ratio corresponding to an increase in the unemployment equivalent to the 5.6 percentage points observed during the 2009 recession, using the coefficients derived from Table 4.¹⁷

The response patterns to having experienced a high unemployment rate in a state during each individual's high school graduation differ significantly across income groups. For those from low-income families, having experienced high unemployment rates around high school graduation is correlated with a low rise in their probability of having attended college and a decline in their probability of having earned a college degree. For high-income individuals the rise is substantially larger both in their likelihoods of having attended and completed college. The implication of these findings is that periods of high unemployment rates around high school graduation are associated with considerable divergence of existing educational composition disparities between low and high-income individuals.

Tables A12 and A13 reveal that the counter-cyclicalities in college enrollment are more pronounced among the male population, being 50% stronger for middle-income and 12% stronger for high-income individuals. Low-income individuals, both male and female, are significantly less responsive. High-income males exhibit a 13% stronger response in college completion rates to changes in the unemployment rate compared to high-income females. This heightened counter-cyclicalities in male enrollment also leads to a higher probability of college dropout following a recession around high school graduation. Specifically, middle-income males react 35% more strongly than middle-income females, and high-income males react 21% more strongly compared to high-income females.

The dynamics of the college dropouts is particularly interesting. A rise in the unemployment rate in the respondent's state concurrent with high school graduation is associated with a subsequent rise in the likelihood of individuals becoming college dropouts relative to those with only a high school diploma. This can be attributed to the pronounced rise in college enrollments subsequent to a decline in the opportunity cost coming from a worse labor market. This surge in enrollment encompasses individuals who, under different economic circumstances, might not have pursued higher education. Consequently, a subset of these individuals, after realizing they are a bad fit for college, opt to drop out from college. As such, periods of high unemployment rates around the time of high school graduation not only impact the overall magnitude of college attendees but also the socio-economic composition of this cohort. The economic downturn encourages individuals

¹⁷Table A11 extends this analysis to associate degrees, focusing on both academic and occupational degrees. The results for occupational degrees show minimal reaction to changes in the unemployment rate. However, for academic degrees, the reaction closely mirrors what we observe for regular college degrees.

who may have not previously contemplated attending college to do so, which eventually heightens the likelihood of observing an augmented count of college dropouts. This effect is especially larger for high-income individuals. Among low-income individuals, the increase in the probability of becoming a college dropout is of a comparatively modest magnitude.

Taken together, the two empirical findings show that the timing of the unemployment shock matters for which income group drives the rise in college dropouts. When the shock hits while enrolled in college, low-income dropouts increase while middle- and high-income students persist. When the shock coincides with high school graduation, middle- and high-income enrollment surges bring in marginal students who subsequently drop out, while low-income individuals are partly precluded from enrolling in the first place.

2.3 Robustness

The main empirical findings are robust to several alternative specifications. For the transitions analysis in Section 2.1, results hold when replacing the contemporaneous unemployment rate with its one-month lag (Table A5), the maximum rate over the preceding four months (Table A7), and the four-month average (Table A8). For the enrollment and completion analysis in Section 2.2, results are unchanged when using the maximum unemployment rate between January and May of the graduation year (Tables A14–A15), the May-only rate as a proxy for graduation-month conditions (Tables A16–A17), and when excluding the COVID years 2020–2021 (Tables A18–A20). Finally, restricting the sample to individuals aged 23 or younger — where the correlation between own earnings and family income is negligible — leaves all main results essentially unchanged (Tables A21–A23).

Tables A24 and A25 add log average published in-state tuition at public four-year institutions (College Board Trends in College Pricing) and log state need-based grant aid per undergraduate FTE (NASSGAP Annual Survey) as controls, restricting both samples to 2004 onward due to data availability, with tuition assigned by academic year. The income heterogeneity in both empirical findings is preserved: the interaction coefficients for middle- and high-income individuals in the dropout and transitions specifications are virtually unchanged from the baseline, the high-income degree completion and dropout responses remain large and statistically significant, and the low-income graduation coefficient is unaffected. The change in sign of the Test Middle and Test High rows in Table A24 reflects a larger baseline coefficient $\hat{\alpha}_1$ in the shorter post-2004 sample rather than any change in the income gradient. State need-based grant aid is statistically insignificant or only marginally significant across specifications and does not alter the main findings.

Finally, Tables A26 and A27 show that both empirical findings are robust to including state-specific linear time trends, which absorb any unobserved state-level policy changes or secular trends that may be correlated with state unemployment rates.

2.4 Discussion of the Main Assumptions

A fundamental assumption underpinning the empirical framework is the appropriateness of using the variable FAMINC from the IPUMS CPS dataset as a reliable proxy for family income, which remains independent of own earnings. Failure to meet this assumption could result in simultaneity bias, wherein higher income individuals are not only more likely to attend college, but individuals who attend college may also have higher earnings following graduation. This assumption gains importance due to the data’s nature, as the information pertaining to family income is collected during interviews, not at the time of high school graduation. Consequently, observations encompass individuals aged around 30 to 40 years who have already graduated from college and currently earn above-average incomes. This situation could introduce bias, suggesting a positive relationship between higher family income and the likelihood of college attendance. While the FAMINC variable includes own earnings, all empirical analyses focus exclusively on young individuals aged 25 years or younger.

Figure A.5 portrays average real earnings by age and family income groups. Starting at age 23, the earnings lines diverge, with higher family income groups earning more on average. However, below the age of 23, there is little variation in own earnings across income groups. In order to increase sample size, particularly for college graduates who might require additional years, all empirical analyses are conducted up to age 25. I performed additional regressions, restricting the sample to ages 23 or younger, and the results remain mainly robust. The results in Table A21 remain consistent when limiting the age to 23 years old. Tables A22 and A23 depict nearly identical findings concerning college enrollment and completion’s counter-cyclical nature across income groups. Consequently, the main empirical findings of the paper hold under scenarios where family income exhibits little to no correlation with own earnings (ages 23 or lower) or is almost uncorrelated (ages 25 or lower).

An additional consideration regarding this variable relates to the family income group observed for each individual, which corresponds to the time of the survey rather than the high school graduation period. A potential concern arises if, during the period from high school graduation to the interview date (spanning up to 7 years or 5 years for the 23-year-old sample), individuals experiencing higher unemployment rates around high school graduation dates are more or less likely to switch their family income category before the interview date. This becomes particularly relevant as estimators could be biased if the unemployment rate in the respondent’s state around high school graduation correlates with the probability of switching income groups during this time frame.

A further identification concern relates to internal migration. The variable $u_{i,s}^{HS}$ is measured in the respondent’s current state of residence, which may differ from the state in which the individual actually graduated from high school if she subsequently migrated. This concern does not apply to the transitions analysis in Section 2.1, where the dependent variable is a month-to-month transition observed within a four-month window and migration between consecutive monthly interviews is negligible. For the cross-sectional analysis in Section 2.2, the concern is more relevant but is bounded by three considerations. First, the sample is restricted to individuals aged 18–25, limiting

the post-graduation window during which migration can occur; the age ≤ 23 robustness check in Tables A22–A23 reduces this window further with negligible effect on the results. Second, migration bias works in opposite directions across income groups: it attenuates the positive high-income coefficient (mobile graduates move to better states, understating their graduation-state shock) while potentially inflating the negative low-income coefficient (dropouts who remain in high-unemployment states overstate their exposure). The high-income result is therefore conservative with respect to migration. Third, the results are stable across specifications with and without state fixed effects, which absorb time-invariant cross-state differences in migration propensity.

The two empirical findings documented in this section establish that recessions generate heterogeneous educational responses across income groups, with low-income individuals disproportionately exposed to both financial dropout risk and adverse labor market entry conditions, while middle- and high-income individuals behave strategically. However, the empirical analysis cannot quantify the lifetime welfare consequences of these educational responses, nor can it decompose the relative importance of the two underlying forces — liquidity constraints and labor market entry scarring — in driving the observed patterns. In Section 3, I introduce a dynamic life-cycle model with endogenous schooling decisions, liquidity constraints, and persistent entry effects that formalizes these tradeoffs and uses the empirical moments documented above to quantify the lifetime costs of experiencing a recession at these two critical periods.

3 Model

I build a dynamic life-cycle individual decision-making model to quantify the lifetime impact of experiencing a rise in the unemployment rate in a state similar to the one experienced in the 2009 recession at two critical points in time for young individuals—high school graduation and college enrollment—across different income levels.

The model incorporates an exogenous productivity process that induces business cycles, differential initial asset levels leading to initial inequalities, liquidity constraints and the endogenous decision to enroll in college and to decide whether to complete or drop out. The model presents tradeoffs. In expansions, individuals are incentivized to enter the labor market to earn high wages. This effect is amplified by labor market rigidities, which cause initial entry conditions to have persistent impacts on future earnings. Conversely, during recessions individuals with sufficient liquidity have strong incentives to enroll in or complete their college education.

Environment

Time (τ) is discrete and finite ($\tau = 0, 1, \dots, \mathcal{T}$).¹⁸ There is no production. Individuals derive utility (U) from consumption (c):

$$U = \mathbb{E}_0 \sum_{\tau=0}^{\mathcal{T}} \beta^{\tau} u(c_{\tau}). \quad (5)$$

Agents know their initial level of assets (a_0), their ex-ante probability of being a good fit for college (p) and the aggregate productivity state of the economy (z_{τ}). Agents choose enrolling ($e = 1$ means to enroll and $e = 0$ means not to enroll) and dropout decisions ($d = 1$ means to drop out and $d = 0$ means not to drop out), which will determine their schooling level (X_{τ}). While enrolled, they will also face an exogenous revelation of their own type. They also choose their consumption (c_{τ}) and next period asset level ($a_{\tau+1}$). The budget constraint that individuals face is the following:

$$(1 + r)a_{\tau} + y_{\tau}(X_{\tau}, z_{\tau}, \tilde{z}) + g(z_{\tau}) = a_{\tau+1} + c_{\tau} + f_{\tau}(X_{\tau}), \quad (6)$$

where

$$a_{\tau+1} \geq \xi. \quad (7)$$

The left-hand side of the budget constraint represents all the income sources available to the individual. a_{τ} is the asset level at period τ , and r is the exogenous rate of return. a_0 can be interpreted as the initial asset holdings of a young individual, mirroring the parental transfers received or at their disposal. Labor income, denoted as y_{τ} , depends on the schooling level X_{τ} , current productivity state z_{τ} , and the productivity state at labor market entry, denoted by $\tilde{z}$. Additionally, the function $g(z_{\tau})$ represents the other type of parental transfers received by the individual that only depend on the current productivity state.¹⁹ This is to ensure that enrolled students remain susceptible to current economic conditions. For example, a negative shock on parental transfers might emulate parental job loss, exerting some financial pressure on the student. The right-hand side shows the various expenditures. $a_{\tau+1}$ represents the asset level in the next period, while c_{τ} is the consumption level. f_{τ} denotes tuition costs, which depend on the schooling level. The asset level for the next period ($a_{\tau+1}$) must not fall below the threshold parameter ξ , embodying liquidity constraints.

The stochastic element z_{τ} corresponds to a productivity shock, and its evolution follows a Markov process $\Pi = [\pi_{ij}]$, where $\pi_{ij} = Pr(z_{\tau+1} = z_j | z_{\tau} = z_i)$. The productivity process follows an

¹⁸I use the subscript τ in the model section instead of t to distinguish it from the empirical section, where t denotes months. In the model, each period τ corresponds to two years.

¹⁹We can think of this as the portion of parental transfers that only depend on the business cycle and maybe initial assets a_0 as the rest of the transfers that depend on the family characteristics besides the cycle. An easy extension of the model could be to make the parental transfers function depend on the interaction between the business cycle and parental assets.

AR(1) process:

$$z_\tau = \mu + \rho z_{\tau-1} + \epsilon_\tau, \quad (8)$$

where ρ is the persistence parameter and ϵ_τ is distributed normally with mean 0 and variance σ^2 .

Schooling levels will be determined by schooling levels in the previous period and by enrollment and dropping out decisions such that:

$$X_{\tau+1} = \Psi(X_\tau, e, d), \quad (9)$$

Timing and Income Structure

The world starts in $\tau = 0$ and the individual realizes the following: her ex-ante probability of being a good fit for college (p), her initial asset level (a_0), and the current economic state (z_0). She also knows that productivity follows a Markov process and the transition probabilities. The individual confronts pivotal decisions regarding schooling, consumption, and savings. During enrollment, the individual does not receive any labor income but she has to pay tuition costs. It takes two periods to complete college. After finishing the first period of enrollment, the individual realizes her aptitude for college, for example, by observing her own grades. The individual only earns the college wage premium if she graduates from college after having observed she is a good fit for it; conversely, an unfavorable fit leaves her with a smaller premium, the college dropout wage premium, even upon successful completion. Subsequently, the individual must opt to either persist in college for the senior year or exit to the labor force in the ensuing period. Once an individual drops out of college she cannot enroll again.

In period $\tau = 0$, the agent must choose whether to enroll in college. If she decides not to enroll, she will be in state $X_0 = N$ and will reconsider her enrollment decision (e) in $\tau = 1$. If the individual enrolls, she will be in state $X_0 = E^+$ or $X_0 = E^-$, depending on the exogenous realization of her suitability for college, such as observing her own grades. The agent knows her probability of being a good fit for college (p). While enrolled (E^+ or E^-), the individual must decide whether to drop out of college (d). Regardless of her type, if she decides to drop out, she will enter state $X_1 = D$ and remain there permanently. If she decides to stay in college, she will enter state $X_1 = S$, which represents the senior year of college. Being a senior in $\tau = 1$ means that the agent will automatically become a college degree holder in $\tau = 2$ when she enters the labor market ($X_2 = C$). However, the individual will only enjoy the college wage premium ($\theta(X_\tau) = \theta(C)$) if she graduates and also receives a favorable realization about her type (E^+). In case of observing a negative type (E^-) she would receive the college dropout wage premium ($\theta(X_\tau) = \theta(D)$ where $\theta(N) < \theta(D) < \theta(C)$) regardless of her decision to complete the degree. I summarize this dynamic evolution of schooling decisions in Figure A.1.

Labor income is a function of schooling state X_τ :

$$y_\tau(X_\tau, z_\tau, \tilde{z}) = \begin{cases} \theta(X_\tau)[\psi_\tau(\exp(\tilde{z})) + (1 - \psi_\tau)(\exp(z_\tau))] & \text{if } X_\tau \in \{N, D, C\} \\ 0 & \text{if } X_\tau \in \{E^+, E^-, S\} \end{cases} \quad (10)$$

$$\theta(X_\tau) = \begin{cases} \theta(C) & \text{if } X_\tau \in \{C\} \text{ and } X_{\tau-\delta} = E^+ \text{ for some } \delta \geq 2, \\ \theta(D) & \text{if } X_\tau \in \{C\} \text{ and } X_{\tau-\delta} = E^- \text{ for some } \delta \geq 2, \\ \theta(D) & \text{if } X_\tau \in \{D\}, \\ \theta(N) & \text{if } X_\tau \in \{N\}. \end{cases}$$

While enrolled in college, labor income is equal to zero. In the other states, income depends on $\theta(X_\tau)$, denoting the college wage premium. Specifically, if an individual has successfully completed college following a favorable assessment of her college suitability, $\theta(X_\tau) = \theta(C)$, where $\theta(C) > \theta(N)$. If the individual drops out of college or receives an unfavorable assessment of her college suitability, she will get $\theta(X_\tau) = \theta(D)$. Moreover, labor income is influenced by the current state of the economy, denoted by z_τ , as well as the economic conditions at the time of labor market entry, represented by $\tilde{z}$. The parameter $0 < \psi_\tau < 1$ governs the extent to which initial conditions impact current income levels and I include a subscript τ to allow for the possibility of shutting down the effect of initial conditions after some τ periods since this effect does not seem to be permanent in the data.

The model generates three types of college dropouts. First, academic dropouts, individuals who, after learning their unsuitability for college (E^-), perhaps due to academic struggles or course failures, opt to drop out of college. In that case, even attaining the degree would not enhance their future θ , making the future investment in college tuition less worth it. This type of dropouts is the type we see in Stange (2012). Second, financial dropouts, individuals who, irrespective of their realization regarding their aptitude for college, confront an adverse productivity shock that disrupts parental transfers ($g(z_\tau)$), plunging them into financial distress and unable to pay for tuition. This mechanism aims to capture the wealth effect of economic recessions. Lastly, strategic dropouts, individuals who, despite realizing their prospective benefits of the college wage premium (E^+), and despite not being financially constrained, choose to abandon college due to an exceptionally robust state of the economy, preferring immediate entry into the labor force. I do not see these types of dropouts occurring in the data nor in the model in the calibration used.

Tradeoffs and Predictions of the Model

The expected income for a recent high school graduate who chooses to enroll in college in $\tau = 0$ is the following:

$$\begin{aligned}
Y^C = & -f + p \left(-f + \theta(C) \mathbb{E}_0 \sum_{\tau=2}^{\tau} [\psi_{\tau}(\exp(z_2)) + (1 - \psi_{\tau})(\exp(z_{\tau}))] \right) \\
& + (1 - p) \left(\theta(D) \mathbb{E}_0 \sum_{\tau=1}^{\tau} [\psi_{\tau}(\exp(z_1)) + (1 - \psi_{\tau})(\exp(z_{\tau}))] \right),
\end{aligned} \tag{11}$$

where $\theta(C)$ represents the college wage premium for a graduate who has realized she is a good fit for college (E^+), thus $\theta(C) > \theta(D) > \theta(N) = 1$. I exclude all parental transfers that depend on the state of the economy ($g(z_{\tau})$) and focus only on labor income net of tuition fees. In this scenario, with probability p , the agent will be a good fit for college, pay tuition for two periods, and enter the labor market in $\tau = 2$, making her $\tilde{z} = z_2$. Conversely, with probability $(1 - p)$, she will not be a good fit for college, drop out, and enter the labor market in $\tau = 1$, resulting in her $\tilde{z} = z_1$.²⁰

The expected income for a recent high school graduate who chooses not to enroll in college is the following:

$$Y^N = \theta(N) \mathbb{E}_0 \sum_{\tau=0}^{\tau} [\psi_{\tau}(\exp(z_0)) + (1 - \psi_{\tau})(\exp(z_{\tau}))]. \tag{12}$$

Higher tuition costs (f) make the choice of attending college less attractive compared to entering the labor market directly. Conversely, higher probabilities of being a good fit for college (p) and a higher college wage premium (θ) increase the attractiveness of pursuing higher education.

Let us focus on the role of business cycles and expectations. A favorable current state of the economy (z_0) makes entering the labor market more appealing, as individuals do not earn wages while enrolled in college. This effect is further amplified by a larger ψ , indicating more persistent labor market entry conditions. Conversely, poor current economic conditions and positive future prospects make attending college more attractive, with these effects also being enhanced by a larger ψ .

Equations (11) and (12) imply that there are some thresholds for p and $\theta(C)$ for which an individual is indifferent between choosing to attend college or entering the labor force.

$$\hat{p} = \frac{\theta(N)\Lambda^0 + f - \theta(D)\Lambda^1}{\theta(C)\Lambda^2 - \theta(D)\Lambda^1 - f}, \tag{13}$$

and

$$\hat{\theta}(C) = \frac{\theta(N)\Lambda^0 - (1 - p)\theta(D)\Lambda^1 + (1 + p)f}{p\Lambda^2}, \tag{14}$$

where $\hat{p}$ corresponds to the ex-ante probability of being a good fit for college such that $Y^C = Y^N$

²⁰In the model I do not force individuals who realize they are a bad fit to drop out of college. However, they always do, and in this example I assume they do.

and $\hat{\theta}(C)$ is the college wage premium such that $Y^C = Y^N$. Λ^2 , Λ^1 and Λ^0 are the lifetime expected income of entering the labor force at $\tau = 2$, $\tau = 1$ and $\tau = 0$, respectively.²¹ I derive both $\hat{p}$ and $\hat{\theta}(C)$ in the Mathematical Appendix B.1.

The model predicts that higher tuition fees (f) would increase both the threshold probability ($\hat{p}$) and the threshold college wage premium ($\hat{\theta}(C)$). This implies that if college becomes more costly, only individuals with a higher *a priori* probability of being a good fit for college would enroll. Additionally, individuals would require a higher college wage premium to be indifferent between attending college and entering the labor market directly.

Higher expected gains from entering the labor force in $\tau = 2$ (Λ^2), for individuals who have spent the first two periods in college, would lower the threshold probability ($\hat{p}$). This means that even individuals less fit for college would be incentivized to enroll, and the required college wage premium for indifference would decrease. Conversely, higher expected lifetime income for individuals entering the labor force in $\tau = 0$ (Λ^0) would increase the threshold probability ($\hat{p}$), making less academically fit individuals prefer not to attend college, and would also raise the required college wage premium ($\hat{\theta}(C)$). The effects of higher expected lifetime income for entering the labor force in $\tau = 1$ are less straightforward.²² The effects on the college wage premium threshold are negative. Higher expected income for entering the labor market in $\tau = 1$ without a college degree decreases the required college wage premium to be indifferent between going to college or not.

In the next section, I describe the model's solution method and calibration strategy before showing the main results.

4 Solution Method and Calibration

To solve the model, I assume individuals exclusively choose their schooling, consumption, and savings during the initial three periods ($\tau = 0$, $\tau = 1$ and $\tau = 2$). From $\tau = 3$ to $\tau = \mathcal{T}$ they only consume a known amount of deterministic income that also depends on their educational status achieved. Each τ in the model is equivalent to 2 years and the world starts when individuals graduate high school, so when they are 18 years old.

I discretize the state space that includes the ex-ante probability of being well-suited for college (p), the initial asset level (a_0), and the initial productivity state (z_0). The utility function is logarithmic ($u(c) = \ln(c)$). I calculate the expected lifetime consumption, earnings, and utility for all a , p and z grids and all potential educational decisions. The feasible educational choices are: never going to college, enrolling in college in the initial period ($\tau = 0$), and enrolling in college in the subsequent period ($\tau = 1$). Enrolling in college involves an upfront tuition cost, denoted by f , which students must pay in their first period. In return, they expect to benefit from a future college wage premium, with probability p of being a good fit for college. If they are indeed a good

²¹ $\Lambda^2 \equiv \mathbb{E}_0 \sum_{t=2}^{\mathcal{T}} [\psi_t \exp(z_2) + (1 - \psi_t) \exp(z_t)]$, $\Lambda^1 \equiv \mathbb{E}_0 \sum_{\tau=1}^{\mathcal{T}} [\psi_\tau \exp(z_1) + (1 - \psi_\tau) \exp(z_\tau)]$ and $\Lambda^0 \equiv \mathbb{E}_0 \sum_{\tau=0}^{\mathcal{T}} [\psi_\tau \exp(z_0) + (1 - \psi_\tau) \exp(z_\tau)]$

²² $\frac{\partial \hat{p}}{\partial \Lambda^1} = \frac{\theta(D)[\theta(N)\Lambda^0 + f - \theta(D)\Lambda^1]}{[\theta(C)\Lambda^2 - \theta(D)\Lambda^1 - f]^2} - \frac{\theta(D)}{\theta(C)\Lambda^2 - \theta(D)\Lambda^1 - f}$

fit, they will continue and pay a second tuition fee f . Otherwise, they will drop out and enter the labor force immediately.

In expected terms, there is no reason to drop out of college, because, if an individual expects to drop out of college in $\tau = 0$ it is not optimal to enroll in college in the first place. College dropouts in the model will only occur if a shock, whether realizing one is a bad fit for college or experiencing a recession that impacts their budget constraints via parental income, consequently pushing the individual into a binding financial situation. I calculate each of these scenarios separately by comparing the utility levels given the optimal asset and consumption trajectories for each case. This sequential process leads to an optimal educational choice, as a function of the individual's probability of a favorable outcome upon college enrollment (p), their initial asset endowment (a), and their initial productivity shock (z). I compute all utilities in expected terms contingent upon the initial productivity shock.

Table 5 presents the calibrated and estimated model parameters. I set some of the parameters in terms of average annual income for no college individuals to get a better intuition of the magnitudes. Given Equation 10 and that each τ is equivalent to 2 years, I use the term $\bar{y}^{NC}$ to refer to the annual average income for no college individuals and $\bar{y}^C = \theta(C)\bar{y}^{NC}$ to refer to the annual average income for college graduates.²³

To match an interest rate of 8% in a span of 2 years, roughly equivalent to a 4% annual rate, I set β accordingly. I fix the time horizon at $\mathcal{T} = 20$, a total of 21 periods after high school graduation ($\tau = 0$), or 42 years (so $18 + 42 = 60$ years old), including $\tau = 0$, where each period denotes a 2-year interval. The part of the parental transfers function that depend only on the state of the economy ($g(z)$) is set to have mean 0, subtracting 1 from z , I used different specifications and the main results hold. The mean of the productivity process (μ) is set to be equal to 1. The autoregressive coefficient ρ is tuned to generate business cycles of 2 years, thereby a recession in the present increases the likelihood of a faster recovery within two years. ξ represents the liquidity constraints in the model and in the baseline model I set it at the 1.5-year average income for non-college individuals $-1.5\bar{y}^{NC}$. I set the college wage premium $\theta(C)$ to match the causal findings by Heckman et al. (2018) and I set the college wage premium for dropouts $\theta(D)$ to be non-existent as with non-college individuals.²⁴ I also present an alternative calibration where I assign a positive wage premium for college dropouts ($\theta(D) > 1$), and the main results are robust. I estimate three parameters using moments in the data. I estimate the variance of the business cycle, the effect of entry conditions on current wages and tuition costs.

To estimate the variance of the business cycle parameter (σ) I target the relative change in real earnings weighted by the probability of being employed for young people during the 2009 crisis. In my sample, real earnings fell by 11.4% but since the probability of being unemployed rises by 5.6

²³The average bi-annual income for no college, given the income process, is equal to $2\bar{y}^{NC} = \exp(\mu)$ so the average annual income is the same divided by 2, $\bar{y}^{NC} = \frac{\exp(\mu)}{2}$. In the calibration, I set μ , the mean of the productivity process equal to 1.

²⁴An alternative version of the model estimates both $\theta(C)$ and $\theta(D)$ to match the difference in lifetime earnings between these two groups and individuals with only a high school diploma as shown in Figure 2. However, these differences are far from causal, that is why in the baseline calibration I used the findings by Heckman et al. (2018).

Table 5: Calibrated and estimated parameters

Parameter	Value	Description
Calibrated:		
β	0.926	2 year discount factor (4% annual int. rate)
a_0^{max}	$16\bar{y}^{NC}$	Maximum starting level of assets
$\mathcal{T}$	20	Total number of 2-y periods (plus $\tau = 0$)
$g(z)$	$6\bar{y}^{NC}(z - 1)$	Cyclical parental transfers mean 0. In recessions falls by $\bar{y}^{NC}$
μ	1	Mean of the productivity process. This gives $\bar{y}^{NC} = \frac{\exp(1)}{2}$
ρ	-0.2	Auto-regressive parameter
ξ	$-1.5\bar{y}^{NC}$	Liquidity constraints. 1.5 year average non-college income
$\theta(C)$	1.2	College wage premium from Heckman et al. (2018)
$\theta(D) = \theta(N)$	1	College wage premium for college dropouts and high-school
Estimated:		
σ^2	0.1623	Variance of the error term of the productivity process
Moment	Emp. Value	Model Value
$\frac{Y^{Rec}}{Y^{Exp}}$	0.834	0.834
ψ	0.12	Effect of entry conditions on wages
Moment	Emp. Value	Model Value
$\frac{Y_{young}^{NC Rec}}{Y_{young}^{NC Exp}}$	0.90	0.89
f	1.09	Tuition fees. Full tuition ($2f$) is equivalent to 160% of $\bar{y}^{NC}$
Moment	Emp. Value	Model Value
$\frac{(\text{Col./HS})^H}{(\text{Col./HS})^L}$	1.94	1.95

Note: Calibrated parameters are set without targeting specific data moments. Estimated parameters are chosen to match the three empirical moments reported in the table. All income-related parameters are expressed in units of $\bar{y}^{NC} = \exp(1)/2$, the average annual income for non-college individuals under $\mu = 1$. Each model period τ corresponds to two calendar years. $\hat{\sigma}^2$ targets the 16.56% weighted earnings drop during the 2009 recession (11.4% earnings decline adjusted for the 5.6 percentage point rise in unemployment probability). $\hat{\psi}$ targets the earnings ratio of non-college individuals entering during a recession versus an expansion over the first six years, taken from Figure 1. $\hat{f}$ targets the college enrollment disparity ratio between high- and low-income young individuals from Table 3.

p.p., the real drop in earnings is of 16.56%. This parameter reflects the size of the cyclical shocks in the model.

The second estimated parameter is the impact of initial labor market conditions on current wages (ψ). This estimation aligns with the earnings of an individual with only a high school diploma, comparing those who entered the labor force during a high unemployment rate period to those who entered during an expansion.²⁵ The empirical values are sourced from Figure 1, focusing on the initial 6 years (3 periods in the model) post labor force entry. The empirical value of 0.899 means that an individual entering the labor force during a recession earns on average 10.10% less than an individual entering in an expansion for the first 6 years.²⁶ I already account for the higher probability of being unemployed during these 6 years.

Lastly, I estimate the parameter f , governing tuition costs. I match this parameter with a moment of the data that displays the disparity in college enrollment between high-income and low-income individuals, shown in the second empirical finding in Section 2.2, more specifically in Table 3. This value for tuition costs, taking into account that full tuition costs are $2f$, corresponds to the 160% of the average annual income for no college ($\bar{y}^{NC}$). Tuition is the primary financial barrier separating enrollment decisions between high- and low-income individuals in the model. The final two columns show the difference between the model and empirical moment values.

5 Results

In this section, I present the main results of the model given the aforementioned parametrization I used. The results align with the main empirical findings I documented in Section 2.

In Table 6, I show the main results that the model reproduces from the empirical section.

The first result shows the percentage of individuals who already received a good signal about their college suitability by income level that drop out of college in expansions and in recessions. In periods of economic expansion there are no financial dropouts. However, in recessions, 14.3% of the low-income group, the most financially vulnerable, drops out of college due to the negative influence of parental transfers ($g(z)$). This behavior is shown in the data in the first columns of Table 2.

The second result explores the cyclical patterns of college enrollment by income levels. I show the percentage of individuals opting for immediate college enrollment (in $\tau = 0$). Immediate college enrollment decreases during recessions for low-income individuals, stays virtually unchanged for middle-income, and increases significantly for high-income. The stronger counter-cyclicity of college enrollment for high-income individuals is shown in Table 4; however, I did not observe this strong pro-cyclicity in college enrollment for low-income in the data. I only observed small

²⁵The reason I focus only on individuals with only a high school diploma is because their earnings and high school graduation are a better proxy for labor market entry and as shown in Figure A.3 college degree holders tend to escape these scarring effects.

²⁶I use a rise in the unemployment rate of the magnitude experienced in 2009, which corresponds to a 5.6 percentage point rise. I also use 6 years instead of 10 due to the fact that in the model there are only 3 periods of endogenous decisions, which correspond to 6 years in the data.

Table 6: Summary of main model results

Income level	Expansion ($z = z_{high}$)			Recession ($z = z_{low}$)		
	Low	Middle	High	Low	Middle	High
Result 1: Financial Dropouts						
% of E^+ who drop out	0	0	0	14.3	0	0
Δ w.r.t expansion				+14.3	0	0
Result 2: Cyclicity of immediate college enrollment						
% enrolling at $\tau = 0$ enrollment	50.9	82	63.2	32.6	81	80.8
Δ w.r.t expansion				-18.3	-1	+17.6
Result 3: Immediate enrollment probability p threshold						
Min p to enroll at $\tau = 0$	50.6	18.8	21.7	68.5	19.8	17.5
Δ w.r.t expansion				+17.9	+1	-4.2
Result 4: Strategic delays						
% enrolling at $\tau = 1$ enrollment	0	0	16	0	0	2.4
Δ w.r.t expansion				0	0	-13.6

Note: Income groups correspond to the initial asset grid: low-income covers grids 1–7 (bottom 35%), middle-income covers grids 8–15 (middle 40%), and high-income covers grids 16–20 (top 25%), mirroring the definitions in the empirical section. All results are averages over the 25 ex-ante college suitability probability grids (p) and all asset grids within each income group. Expansion and recession refer to the high and low productivity states (z_{high} and z_{low}) respectively. The Δ w.r.t. expansion rows (in bold) report the difference between the recession and expansion values for each result, making the direction and magnitude of cyclicity directly readable across income groups.

Result 1 reports the share of enrolled E^+ students — those who have already received a favorable suitability signal — who drop out due to a binding liquidity constraint. These are financial dropouts, as distinct from academic dropouts (E^- students choosing to leave) and strategic dropouts (financially unconstrained agents who exit due to a strong labor market). **Result 2** reports the fraction of agents who enroll in college immediately at $\tau = 0$. **Result 3** reports the minimum ex-ante probability of being a good fit for college (p) required for an agent to find immediate enrollment optimal, averaged across asset levels within each income group. Higher thresholds in recession indicate that only more academically suited individuals find enrollment worthwhile when economic conditions are unfavorable. **Result 4** reports the fraction of agents who delay college enrollment to $\tau = 1$ having not enrolled at $\tau = 0$, reflecting the option value of observing improved labor market conditions before committing to college.

counter-cyclicality instead. This discrepancy between model and data for low-income enrollment reflects that the financial pressure that drives pro-cyclical dropout while enrolled simultaneously discourages initial enrollment around high school graduation, since the same liquidity constraints and parental income shocks operate through both margins. The model does, however, correctly capture two important dimensions of the data: the full pattern of graduation cyclicality while enrolled and the relative ordering across income groups around high school graduation.

The third result shows the ex-ante probability thresholds for each income group deciding to immediately enroll in college. For low-income, the p threshold rises significantly during recessions, that is, only individuals who are *a priori* more likely to succeed in college enroll. This p almost does not change for middle income, but it does drop significantly for the high-income group. This fall in p will result in a subsequent rise in academic dropouts, as we observe in the data in Table 4 coming mainly from the high-income group. In the data, I also observed this type of college dropouts, although in a significantly smaller scale, with lower- and middle-income individuals.

Finally, I show how the incentives to strategically delay labor market entry to avoid the scarring effects generated by economic recessions vary by income level. I only observe this behavior from the high-income group, especially high-income individuals during expansions, who defer college entry to capitalize on the favorable labor market conditions. This aligns with the trends shown in the last two columns of Table 2.

Table A28 reports results from an alternative calibration in which the college dropout wage premium exceeds one ($\theta(D) > 1$). As expected, overall college enrollment increases, since individuals who drop out still earn more than high-school graduates. Nevertheless, the main results are essentially unchanged. Table A29 presents an additional calibration allowing the college wage premium to vary across income groups, with higher initial asset levels associated with higher premiums. Once again, the qualitative results remain unchanged.

6 Counterfactuals

In this section, I present the two main counterfactuals of the paper, which answer the two main research questions of the paper: What is the lifetime impact of experiencing an economic recession while being enrolled in college by income levels? And what is the lifetime impact of experiencing an economic recession at high school graduation by different income levels? I will also address the same simulations but exploring the role of the two main rigidities in the model: liquidity constraints and the persistent effects of labor market entry conditions on wages.

First, I quantify the lifetime impacts of experiencing an economic recession while enrolled in college by income levels. For this first counterfactual I only consider enrolled students who already observed they are a good fit for college and therefore if they decide to complete the degree they will enjoy college wage premium. Second, I quantify the lifetime impacts of experiencing an economic recession around high school graduation by income levels by averaging out all grids of p , the *ex-ante* probability of being a good fit for college. For the graph below I fix a p for illustration purposes

while maintaining the same 2-D characteristics.²⁷ In Figure 3, I plot the expected lifetime utility levels for each initial asset level. The blue curve represents an individual experiencing an economic expansion during their college enrollment or around high school graduation, while the red curve represents an individual experiencing a recession during the same period.

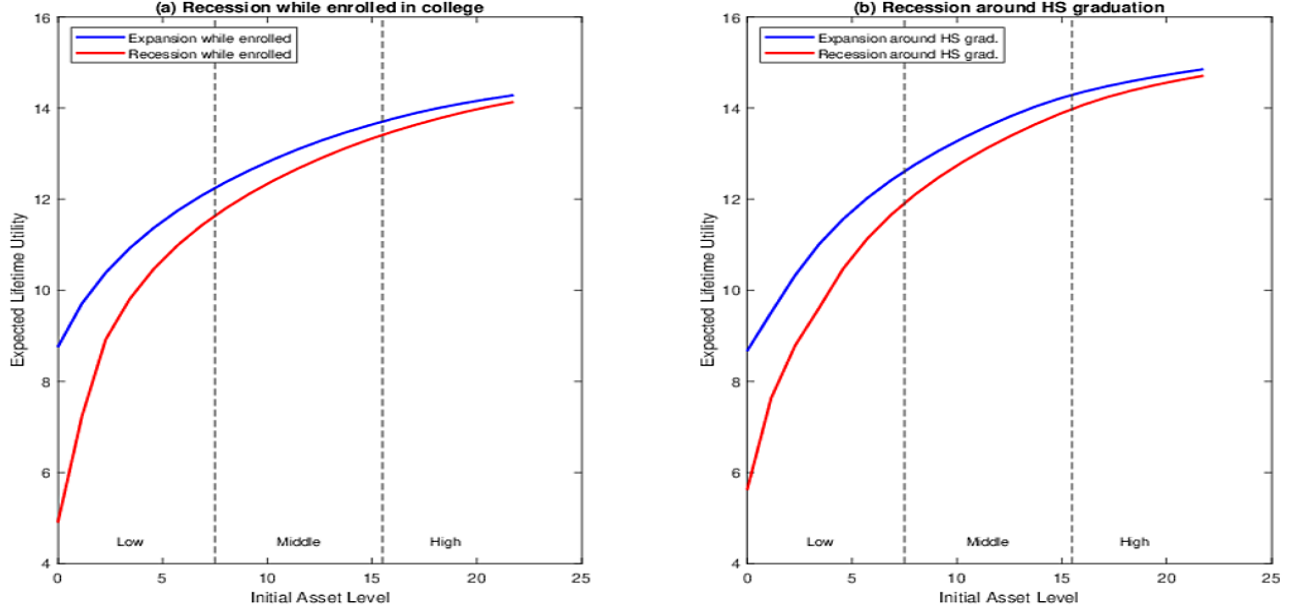


Figure 3: Lifetime effects of experiencing an economic recession by income level

Note: Each panel plots expected lifetime utility as a function of the initial asset level (x-axis, ranging from grid 1 = lowest assets to grid 20 = highest assets) under expansion (blue) and recession (red). Dashed vertical lines mark the income group boundaries: low-income corresponds to asset grids 1–7 (bottom 35% of the asset distribution), middle-income to grids 8–15 (middle 40%), and high-income to grids 16–20 (top 25%), mirroring the empirical definitions. Panel (a) fixes the initial productivity state to a recession while the individual is already enrolled in college and has received a favorable type realization (E^+), isolating the financial dropout channel. Panel (b) fixes the initial productivity state to a recession at the time of high school graduation, averaging across all 25 ex-ante college suitability probability grids (p), with results illustrated for $p = 62.5\%$ (the 16th grid out of 25) for expositional clarity. In both panels, the gap between the blue and red curves represents the lifetime utility loss from experiencing a recession rather than an expansion at that critical juncture, and is largest and most visible for low-income individuals.

Experiencing an economic recession has negligible effects across all income groups except for the low-income one. Among low-income individuals, the expected loss reaches 15% in lifetime utility and 8.8% in present value consumption when the shock occurs while enrolled in college, and 13.8% in utility and 8.5% in present value consumption when it occurs around high school graduation. For middle-income individuals the corresponding losses are 3.1% and 5.6% in utility and present value consumption respectively, for the first counterfactual, and 3.4% and 5.7% for the second. High-

²⁷In the baseline scenario I fix p to be equal to 62.5% which corresponds to the 16th grid out of 25. I also explore how these losses depend on p .

income individuals suffer the smallest losses, at 1.4% in utility and 4.7% in consumption for the first counterfactual and 1.5% and 4.4% for the second. The gap between utility and consumption losses reflects the logarithmic properties of the utility function: log utility amplifies losses at low consumption levels and compresses them at high levels, widening the gap between income groups in utility space relative to consumption space.

Low-income individuals are the most affected. If the economic shock occurs while they are enrolled in college, 14% drop out and enter the labor market during unfavorable conditions—missing out on the college wage premium and facing poor initial job prospects. If the shock happens around high school graduation, they are less likely to enroll in college and still face a weak labor market. In contrast, middle- and high-income individuals tend to use college enrollment and completion as a buffer against recessions, and during economic expansions, they often delay college to take advantage of higher wages.

To bridge the empirical and structural sections more directly, I decompose the model-implied recession effect on college graduation into three channels using a Kitagawa decomposition: an enrollment margin, a type composition margin, and a pure financial dropout margin. For low-income individuals, the enrollment channel accounts for 83% of the total graduation cyclicity, the dropout channel for 22%, and the composition channel is -5% , the latter reflecting that in recession the enrolled pool is of higher average quality since the p threshold rises. The dominance of the enrollment channel for low-income is partly a consequence of the model’s known limitation — the procyclical enrollment prediction for low-income — and that empirically the dropout channel likely plays a larger role than the decomposition suggests.

I now explore the role of fixing a different probability of succeeding in college p for the second counterfactual, since the first counterfactual focuses only on individuals who already know they are a good fit for college. I also explore the role of college tuition (f) and the influence of labor market entry conditions (ψ).

In Table 7, I show the % losses from experiencing a recession in both periods of analysis with the baseline parametrization shown in Table 5 and comparing it with other policy-relevant counterfactuals. The first three columns correspond to the effects for students enrolled in college who already learned about their college suitability and the last three columns to the effects for individuals around their high-school graduation. As always, I separate between the three main income groups. In the first experiment I show the different expected lifetime utility losses from experiencing an economic recession for different probabilities of being a good fit for college (p), for this experiment, I only show the effects of the economic recession hitting around high-school graduation since the first counterfactual only focuses on students who already know they are a good fit for it. In the second experiment, I show the effects of making college tuition cheaper and more expensive in the economy and in the last one I show the effects of eliminating and increasing the effects of labor market entry conditions on current wages.

In the first experiment, I show that recessions disproportionately harm low-income individuals, particularly those with a high *ex-ante* probability of succeeding in college. These individuals stand

Table 7: Policy relevance: The role of p , f and ψ

Income level	% Loss in Expected Lifetime Utility in a Recession					
	While enrolled			Around HS graduation		
	Low	Middle	High	Low	Middle	High
1) Prob. of good fit p						
$p = 12.5\%$	-	-	-	13.21	3.39	1.92
Baseline averaging across p	-	-	-	13.83	3.37	1.45
$p = 100\%$	-	-	-	14.78	3.21	1.41
2) Tuition costs f						
$2f = 100\%\bar{y}^{NC}$	13.99	3.01	1.35	13.87	3.18	1.35
Baseline $2f = 160\%\bar{y}^{NC}$	14.97	3.13	1.41	13.83	3.37	1.45
$2f = 200\%\bar{y}^{NC}$	15.48	3.22	1.46	13.76	3.49	1.51
3) Labor market entry conditions ψ						
$\psi = 0$	14.57	3.17	1.43	13.50	3.33	1.42
Baseline $\psi = 0.12$	14.97	3.13	1.41	13.83	3.37	1.45
$\psi = 0.5$	16.62	3.02	1.36	15.09	3.50	1.54

Note: Each entry reports the percentage loss in expected lifetime utility from experiencing a recession rather than an expansion, $(\mathbb{E}[U | z_{low}]/\mathbb{E}[U | z_{high}] - 1) \times 100$, averaged over all p and asset grids within each income group. The first three columns condition on the shock occurring while enrolled with a favorable type realization (E^+); the last three condition on the shock occurring around high school graduation. Experiment 1 is reported only for the around-HS counterfactual since the while-enrolled counterfactual already selects E^+ individuals. In Experiment 2, $2f$ is total tuition expressed as a share of average annual non-college income $\bar{y}^{NC}$. In Experiment 3, $\psi = 0$ implies no entry scarring and $\psi = 0.5$ implies stronger persistence of initial labor market conditions on wages.

to gain the most from college enrollment but are unable to access it due to financial constraints. In contrast, among high-income individuals, recessions tend to affect those with lower probabilities of college success. Economic downturns incentivize them to enroll in college despite their low likelihood of completing it successfully.

In the second experiment, I vary tuition costs f to explore their role in shaping recession losses across income groups. The differential effects of tuition costs across income groups are amplified by the presence of liquidity constraints: lower tuition reduces the probability of financial dropout while enrolled, disproportionately benefiting low-income individuals who are closest to the constraint. Around high school graduation, the effects are asymmetric — as discussed in detail in the previous section — with lower tuition increasing losses for low-income but reducing them for middle- and high-income, because lower barriers expand enrollment in expansion without sufficiently overcoming the recession-induced financial pressure for low-income individuals.

In the final experiment, I show that economies with more rigid labor markets (higher ψ) generate larger incentives for high-income individuals to use college strategically, but also larger scarring losses for low-income dropouts, implying that reducing entry frictions has heterogeneous welfare effects by income. When a recession hits during college, eliminating this friction leads to larger losses for middle and high-income, as they stay enrolled and enter the labor market in the following period, which is likely going to be an expansion. Due to the negative auto-regressive parameter ($\rho = -0.2$), they would benefit more from stronger entry effects to sustain high wages. Thus, greater rigidity reduces losses. For low-income individuals, the effects are mixed: those who remain in college benefit from the rigidity, while those who drop out prefer its absence, as they enter the labor market during the recession. When the recession occurs around high school graduation, the impact of labor market entry rigidities becomes clearer. Intensifying these frictions amplifies losses across all groups, reduces college enrollment, and makes strategic delays among high-income individuals even more pronounced. In Table A30, I present the same experiments but using the % loss in expected present value of consumption instead of utility.

7 Conclusions

College decisions in response to macroeconomic conditions can either amplify or buffer the long-term effects of recessions. Low-income individuals are particularly vulnerable: recessions often force them to drop out of college and enter the labor market under unfavorable conditions, leading to persistent earnings losses. In contrast, high-income individuals use college enrollment and completion strategically to shield themselves from downturns and benefit from better labor market entry.

When recessions occur during college, low-income individuals face average lifetime utility losses of 15%, which worsen under tighter liquidity constraints and stronger labor market rigidities. Middle-income individuals lose around 3%, and high-income only 1.4%, as they tend to remain enrolled and graduate.

If the recession hits around high school graduation, low-income individuals lose 13.8% in lifetime utility, compared to 3.4% for middle-income and 1.5% for high-income groups. These losses are again exacerbated by financial constraints and labor market frictions. Notably, recessions prevent highly prepared low-income individuals from attending college, while incentivizing less prepared high-income individuals to enroll—often resulting in higher dropout rates.

A limitation of the model is its difficulty in simultaneously generating pro-cyclical dropout while enrolled and counter-cyclical enrollment around high school graduation for low-income individuals. The same financial pressure channel that drives low-income students out of college during recessions also discourages initial enrollment, creating a tension that the current model cannot resolve within a single set of parameters. Writing down a model that jointly replicates both empirical patterns for low-income individuals would likely require richer heterogeneity in how liquidity constraints bind across different stages of the educational decision and I think it would be a valuable contribution to the literature on inequality and business cycles.

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A Additional Figures and Tables

Figures

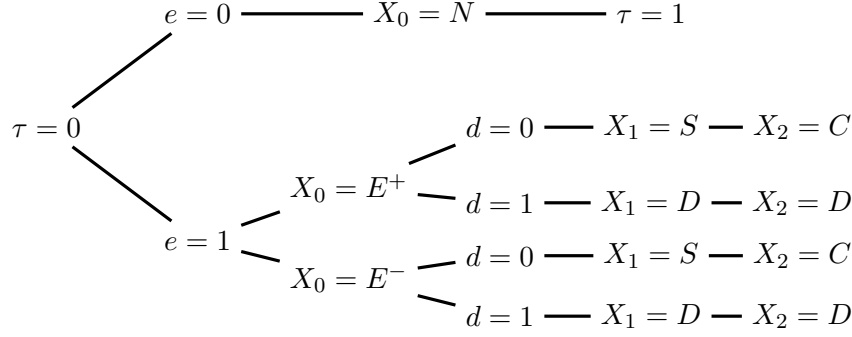


Figure A.1: Diagram of schooling decisions

Note: The diagram shows the sequential schooling decisions in the model. At $\tau = 0$ the agent chooses whether to enroll ($e = 1$) or not ($e = 0$). If she enrolls, she receives a type realization (E^+ or E^-) and decides whether to drop out ($d = 1$) or continue ($d = 0$). States S , D , and C denote senior year, dropout, and college graduate respectively.

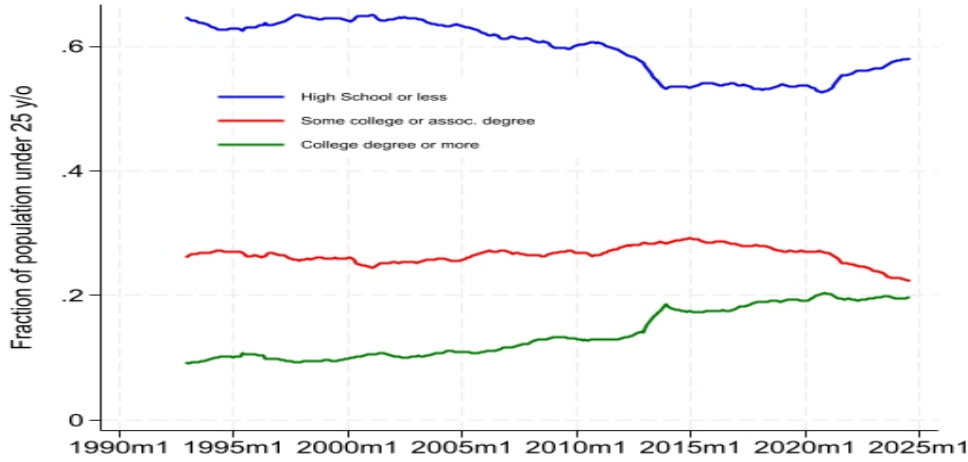


Figure A.2: Evolution of educational groups over time.

Note: Twelve-month moving averages of the fraction of individuals under 25 in each educational group. Currently enrolled students are excluded; fractions sum to 1 in each month. Source: IPUMS CPS, January 1992 to July 2024.

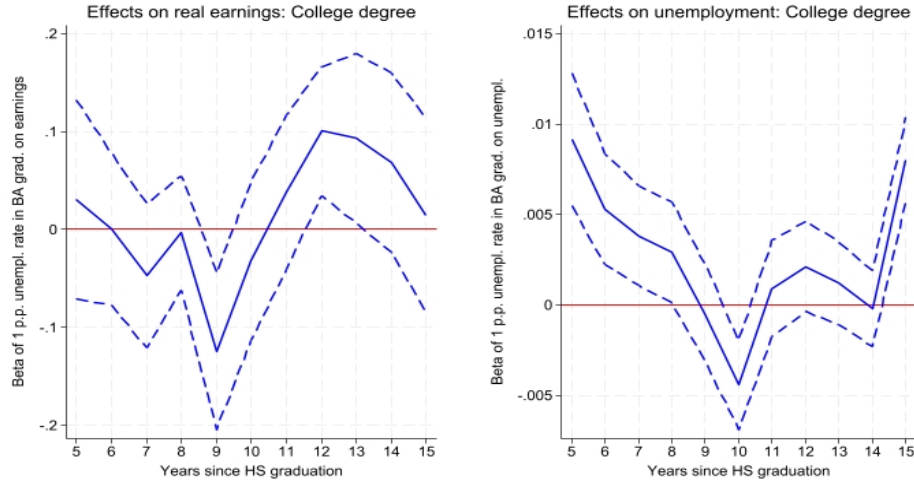


Figure A.3: Negative and persistent effects of graduating from college during recessions.

Note: Plots $\hat{\beta}_1$ by years since high school graduation from regressions of log real earnings and unemployment on u_i^{BA} , the state unemployment rate when the cohort graduated from college (assumed four years after age 18). Sample restricted to individuals with a bachelor's degree only. Dashed lines are 95% confidence intervals. See Section 2.1 for the full specification.

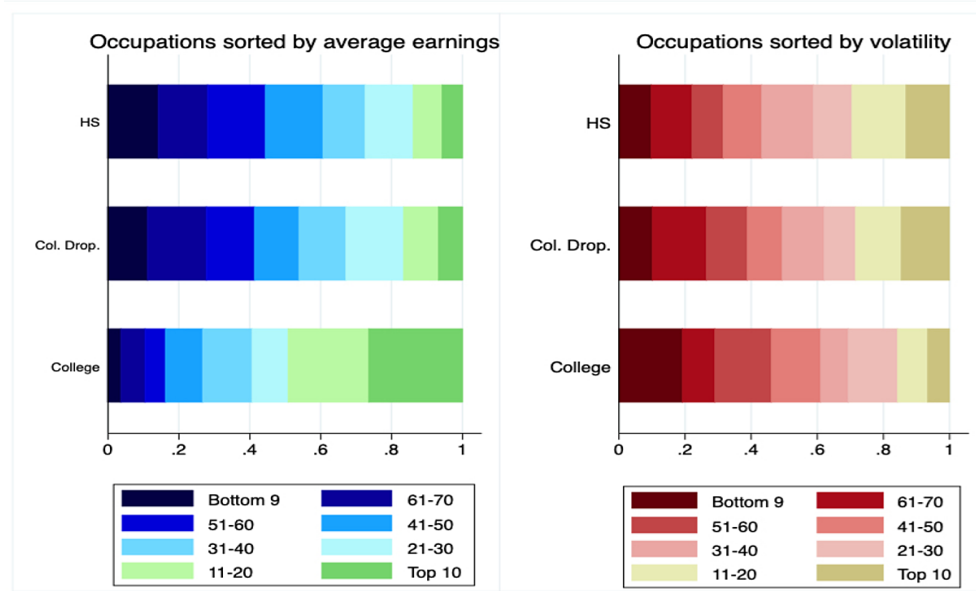


Figure A.4: Occupation sorting by earnings and volatility. Relative proportion by educational group.

Note: Share of workers in each occupation earnings decile (left panel) and volatility decile (right panel) by educational group. Occupations are sorted by average real earnings and earnings volatility respectively across the full sample period. Source: IPUMS CPS.

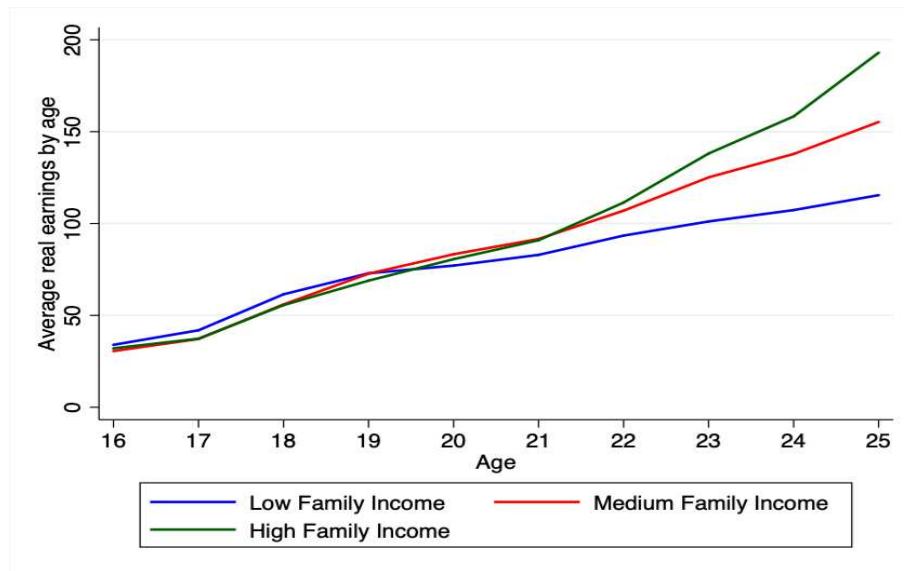


Figure A.5: Differences in real earnings by age and family income group

Note: Mean real earnings in 1992 dollars by single year of age and family income group. Earnings lines begin to diverge at age 23, justifying the restriction to individuals aged 25 or younger in the main empirical analysis. Source: IPUMS CPS.

Table A1: College enrollment is counter-cyclical using linear and exponential trends

	Effect on college enrollment deviations from trend
Unemployment rate (p.p.)	0.398*** (0.077)
Unemployment rate deviations from NAIRU (p.p.)	0.405*** (0.077)
Recession (binary)	0.242 (0.322)
Real GDP growth (YoY%)	-0.233*** (0.064)

Note: Replicates Table 1 adding linear and exponential time trends to the baseline specification. Results are unchanged. Robust standard errors in parentheses. Source: CPS, World Bank, UN population data, Federal Reserve Bank of St. Louis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Tables

Table A2: IPUMS CPS data: Descriptive Statistics weighted

	Longitudinal		Cross-sectional	
	All ages	Age ≤ 25	All ages	Age ≤ 25
Age	30.19	20.09	30.48	20.28
Sex (Female %)	50.42	50.21	50.37	50.14
Race (White %)	75.39	76.11	75.48	76.25
High-school or less (%)	48.35	61.79	47.41	60.22
Some college (%)	27.11	28.48	27.86	29.96
Bachelor's or more (%)	24.54	9.73	24.73	9.81
Weekly earnings (\$)	796.43	416.87	775.26	387.43
Family Income ($x =$ Yearly Income)				
Low % ($x < \$30k$)	26.87	35.67	26.63	35.58
Middle % ($\$30k \leq x < \$75k$)	36.23	36.95	36.18	36.90
High % ($x \geq \$75k$)	36.90	27.38	37.19	27.52
Labor force (%)	72.56	61.61	72.29	61.68
Employed (%)	66.51	53.53	66.83	54.33
Unemployed (%)	6.06	8.08	5.46	7.35
Appearances	3.78	3.74	1	1
Individuals	2,158,404	1,086,525	2,158,558	1,075,005

Note: Descriptive statistics computed using CPS sampling weights (variable `wtfinl`). The longitudinal sample retains up to four consecutive monthly observations per individual. The cross-sectional sample keeps one observation per individual. Income brackets follow the definitions in Section 1.2. Source: IPUMS CPS.

Table A3: Education variable change in criterion

Variable label	Jan. 1976 - Dec. 1991	Jan. 1992 - Dec. 2022
Low		
None, preschool, or kindergarten	✓	✓
Grades 1, 2, 3, or 4	X	✓
Grade 1	✓	X
Grade 2	✓	X
Grade 3	✓	X
Grade 4	✓	X
Grades 5 or 6	X	✓
Grade 5	✓	X
Grade 6	✓	X
Grades 7 or 8	X	✓
Grade 7	✓	X
Grade 8	✓	X
Grade 9	✓	✓
Grade 10	✓	✓
Grade 11	✓	✓
Grade 12th grade, no diploma	X	✓
Grade 12th grade, diploma unclear	✓	X
High school diploma or equivalent	✓	✓
Middle		
1 year of college	✓	X
2 years of college	✓	X
Some college but no degree	X	✓
Associate's degree, occ/voc prog.	X	✓
Associate's degree, academic prog.	X	✓
3 years of college	✓	X
High		
4 years of college	✓	X
Bachelor's degree	X	✓
5 years of college	✓	X
6 years of college	✓	X
Master's degree	X	✓
Professional school degree	X	✓
Doctorate degree	X	✓

Note: The table documents the change in educational category labels in the CPS between the pre-1992 and post-1992 periods. The category "Some college but no degree," essential for identifying college dropouts, is only available from January 1992 onward, motivating the sample start date. Source: IPUMS CPS.

Table A4: Transitions from enrolled to dropout and from labor force to enrolled using pool cross-section

	College Dropouts		LF → College	
$u_{s,t}$	0.143*** (0.033)	0.131* (0.073)	-0.028 (0.022)	0.050 (0.071)
Mid. Inc. $\times u_{s,t}$	-0.236*** (0.048)	-0.192*** (0.070)	0.461*** (0.031)	0.367*** (0.088)
High. Inc. $\times u_{s,t}$	-0.182*** (0.046)	-0.142* (0.079)	0.665*** (0.038)	0.458*** (0.124)
Obs.	706,702	706,702	2,024,651	2,024,651
R-squared	0.001	0.004	0.007	0.015
Controls	No	Yes	No	Yes
Test Middle ($\alpha_1 + \alpha_2$)	-0.009***	-0.060	0.434***	0.417***
Test High ($\alpha_1 + \alpha_3$)	-0.039	-0.011	0.638***	0.508***

Note: Replicates Table 2 using a pooled cross-section instead of a panel random effects estimator. Results are nearly identical. Effects in percentage points relative to staying in college or staying in the labor force. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Transitions from enrolled to dropout and from labor force to enrolled using the lag of unemployment rate

	College Dropouts	LF → College
$u_{s,t-1}$	0.161** (0.066)	-0.119 (0.081)
Mid. Inc. $\times u_{s,t-1}$	-0.190*** (0.062)	0.533*** (0.104)
High. Inc. $\times u_{s,t-1}$	-0.176*** (0.066)	0.645*** (0.134)
Obs.	706,702	2,024,651
R-squared	0.004	0.015
Test Middle ($\alpha_1 + \alpha_2$)	-0.029	0.415***
Test High ($\alpha_1 + \alpha_3$)	-0.015	0.526***

Note: Replicates Table 3 replacing the contemporaneous state unemployment rate with its one-month lag. Results are robust to this alternative timing. Effects in percentage points. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Transitions from enrolled to dropout and from labor force to enrolled by sex

	Dropouts	LF → College	Dropouts	LF → College
	Male		Female	
$u_{s,t}$	0.149** (0.060)	-0.081 (0.075)	0.143* (0.076)	-0.104 (0.093)
Mid. Inc. $\times u_{s,t}$	-0.216*** (0.056)	0.430*** (0.084)	-0.176* (0.093)	0.633*** (0.140)
High. Inc. $\times u_{s,t}$	-0.208*** (0.070)	0.647*** (0.137)	-0.206** (0.086)	0.677*** (0.149)
Obs.	327,825	1,064,289	378,877	960,362
R-squared	0.004	0.010	0.004	0.012
Test Middle ($\alpha_1 + \alpha_2$)	-0.068	0.349***	-0.033	0.529***
Test High ($\alpha_1 + \alpha_3$)	-0.059	0.567***	-0.063**	0.572***

Note: Replicates Table 2 separately for male and female respondents. Results are virtually identical across sexes. Effects in percentage points relative to staying in college or staying in the labor force. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Transitions from enrolled to dropout and from labor force to enrolled using max unemployment rate

	College Dropouts	LF → College
$\max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.140** (0.057)	-0.127* (0.077)
Mid. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	-0.174*** (0.056)	0.478*** (0.095)
High. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	-0.179*** (0.061)	0.537*** (0.120)
Obs.	706,702	2,024,651
R-squared	0.004	0.015
Test Middle ($\alpha_1 + \alpha_2$)	-0.034	0.351***
Test High ($\alpha_1 + \alpha_3$)	-0.038	0.410***

Note: Replicates Table 2 replacing $u_{s,t}$ with the maximum state unemployment rate over the four months prior to and including the interview month. Results are robust. Effects in percentage points. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Transitions from enrolled to dropout and from labor force to enrolled using average unemployment rate

	College Dropouts	LF → College
$\bar{u}_{s,t}$	0.129** (0.059)	-0.099 (0.081)
Mid. Inc. $\times \bar{u}_{s,t}$	-0.191*** (0.059)	0.566*** (0.106)
High. Inc. $\times \bar{u}_{s,t}$	-0.225*** (0.070)	0.733*** (0.145)
Obs.	706,702	2,024,651
R-squared	0.004	0.015
Test Middle ($\alpha_1 + \alpha_2$)	-0.062*	0.467***
Test High ($\alpha_1 + \alpha_3$)	-0.095***	0.634***

Note: Replicates Table 2 replacing $u_{s,t}$ with the average state unemployment rate over the four months prior to and including the interview month. Results are robust. Effects in percentage points. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Composition of education by income groups, individuals aged 21–25

Including enrolled students			
% within income group	HS diploma only	Gone to college	College degree
low-income	34.12	49.26	13.29
middle-income	32.88	59.05	20.30
high-income	22.42	74.00	32.20
Disparity Ratios		College / HS	Degree / HS
Mid. inc. / low-income		1.11	1.59
High inc. / low-income		2.29	3.69
Excluding enrolled students			
% within income group	HS diploma only	Dropout	
low-income	42.28	18.60	
middle-income	39.81	20.18	
high-income	29.74	19.31	
Disparity Ratios		Drop. / HS	
Mid. inc. / low-income		1.15	
High inc. / low-income		1.48	

Note: Replicates Table 3 restricting the sample to individuals aged 21–25 to address the concern that late high school graduation at younger ages may disproportionately affect low-income individuals and distort the disparity ratios. By age 21, late high school graduation is essentially complete. Row shares within each income group do not sum to 100% because the remaining fraction consists of individuals who have not yet completed high school. The bottom panel excludes currently enrolled students to avoid classifying active students as dropouts. Disparity ratios are qualitatively similar to those in Table 3 and if anything larger. All statistics use CPS sampling weights. Source: IPUMS CPS.

Table A10: Implied compositional change by income groups

Disparity Ratios	Col. enrolled / HS		Drop. / HS		Col. degree / HS	
	Avg.	Shock	Avg.	Shock	Avg.	Shock
Mid. inc. / Low inc.	1.19	1.30	1.14	1.22	1.50	1.67
High inc. / Low inc.	1.94	2.16	1.48	1.63	2.81	3.67

Note: Each cell reports the ratio of college-to-HS participation for middle- or high-income individuals relative to the same ratio for low-income. The *Avg.* column uses baseline shares from Table 3; the *Shock* column applies the regression coefficients from Table 4 to a 5.6 percentage point unemployment increase, equivalent to the 2009 recession. Source: IPUMS CPS.

Table A11: Associate degrees and the business cycle

	Assoc. occupational		Assoc. academic	
	(1)	(2)	(3)	(4)
$u_{i,s}^{HS}$	-0.140*** (0.039)	-0.044 (0.052)	0.077* (0.044)	-0.029 (0.067)
Mid. Inc. $\times u_{i,s}^{HS}$	0.137** (0.060)	0.058 (0.079)	0.265*** (0.068)	0.180 (0.122)
High. Inc. $\times u_{i,s}^{HS}$	0.215*** (0.076)	0.018 (0.083)	0.500*** (0.091)	0.318** (0.142)
Controls	No	Yes	No	Yes
Obs.	286,910	286,910	290,301	290,301
R-squared	0.002	0.032	0.006	0.051
F-test Middle ($\beta_1 + \beta_2$)	-0.003	0.013	0.342***	0.152
F-test High ($\beta_1 + \beta_3$)	0.075	-0.026	0.577***	0.290**

Note: Replicates Table 4 replacing the dependent variable with indicators for holding an occupational associate degree (columns 1–2) or an academic associate degree (columns 3–4). Results show that occupational degrees are acyclical while academic degrees mirror the pattern for four-year college degrees. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: College enrollment and completion and the business cycle by sex

	Att. college	College or more	Att. college	College or more
	Male		Female	
$u_{i,s}^{HS}$	0.199 (0.124)	-0.243** (0.112)	0.181 (0.152)	-0.330** (0.140)
Mid. Inc. $\times u_{i,s}^{HS}$	0.461** (0.183)	0.270 (0.163)	0.260 (0.315)	0.287 (0.306)
High. Inc. $\times u_{i,s}^{HS}$	0.386 (0.234)	0.960*** (0.170)	0.340 (0.305)	0.965*** (0.274)
State Controls	Yes	Yes	Yes	Yes
Clustered SE state	Yes	Yes	Yes	Yes
Obs.	317,594	179,901	347,958	184,886
R-squared	0.049	0.224	0.060	0.289
Test Middle ($\beta_1 + \beta_2$)	0.661***	0.027	0.442**	-0.044
Test High ($\beta_1 + \beta_3$)	0.585***	0.717***	0.521**	0.635***

Note: Replicates the enrollment and degree completion columns of Table 4 separately for male and female respondents. Counter- cyclicity is more pronounced among males for middle- income and similar across sexes for high-income. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A13: College dropouts and college degree and the business cycle by sex

	College Dropout	
	Male	Female
$u_{i,s}^{HS}$	0.300** (0.129)	0.220* (0.108)
Mid. Inc. $\times u_{i,s}^{HS}$	0.314 (0.206)	0.235* (0.282)
High. Inc. $\times u_{i,s}^{HS}$	0.457** (0.209)	0.403 (0.332)
State Controls	Yes	Yes
Clustered SE state	Yes	Yes
Obs.	166,521	157,832
R-squared	0.027	0.027
Test Middle ($\beta_1 + \beta_2$)	0.614***	0.455*
Test High ($\beta_1 + \beta_3$)	0.757***	0.624**

Note: Replicates the college dropout column of Table 4 separately for male and female respondents. The counter-cyclical surge in dropouts is stronger among males. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A14: College enrollment and completion and the business cycle using the maximum unemployment between January and May

	Attended college		College Degree or more	
$\max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.047 (0.055)	0.162 (0.103)	-0.173*** (0.055)	-0.194** (0.085)
Mid. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.174** (0.076)	0.090 (0.162)	0.252*** (0.079)	0.125 (0.158)
High. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.184** (0.077)	-0.016 (0.153)	0.582*** (0.095)	0.271** (0.111)
State Controls	No	Yes	No	Yes
Clustered SE state	No	Yes	No	Yes
Obs.	665,552	665,552	364,787	364,787
R-squared	0.019	0.059	0.042	0.262
Test Middle ($\beta_1 + \beta_2$)	0.221***	0.251**	0.079	-0.069
Test High ($\beta_1 + \beta_3$)	0.231***	0.145	0.408***	0.078

Note: Replicates Table 4 replacing $u_{i,s}^{HS}$ with the maximum state unemployment rate between January and May of the graduation year. Results are robust. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A15: College dropouts and college degree and the business cycle using the maximum unemployment between January and May

	College Dropout	
$\max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.020 (0.065)	0.163 (0.103)
Mid. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.115 (0.091)	0.110 (0.174)
High. Inc. $\times \max \{u_{s,t-4}, u_{s,t-3}, u_{s,t-2}, u_{s,t-1}, u_{s,t}\}$	0.217** (0.108)	0.107 (0.138)
State Controls	No	Yes
Clustered SE state	No	Yes
Obs.	324,353	324,353
R-squared	0.005	0.029
Test Middle ($\beta_1 + \beta_2$)	0.135**	0.273*
Test High ($\beta_1 + \beta_3$)	0.237***	0.270***

Note: Replicates the dropout column of Table 4 using the maximum state unemployment rate between January and May of the graduation year. Results are robust. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A16: College enrollment and completion and the business cycle using only May

	Attended college		College Degree or more	
$u_{i,s}^{HS}$	0.085	0.158	-0.145**	-0.253***
	(0.057)	(0.108)	(0.058)	(0.095)
Mid. Inc. $\times u_{i,s}^{HS}$	0.294***	0.183	0.392***	0.191
	(0.079)	(0.183)	(0.084)	(0.181)
High. Inc. $\times u_{i,s}^{HS}$	0.355***	0.088	0.953***	0.463***
	(0.081)	(0.175)	(0.102)	(0.132)
State Controls	No	Yes	No	Yes
Clustered SE state	No	Yes	No	Yes
Obs.	665,552	665,552	364,787	364,787
R-squared	0.019	0.059	0.042	0.263
Test Middle ($\beta_1 + \beta_2$)	0.379***	0.340***	0.248***	-0.063
Test High ($\beta_1 + \beta_3$)	0.441***	0.246**	0.809***	0.209**

Note: Replicates Table 4 replacing $u_{i,s}^{HS}$ with the state unemployment rate in May only of the graduation year, a proxy for conditions at the moment of high school graduation. Results are robust. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A17: College dropouts and college degree and the business cycle using only May

	College Dropout	
$u_{i,s}^{HS}$	0.105	0.204*
	(0.068)	(0.111)
Mid. Inc. $\times u_{i,s}^{HS}$	0.181*	0.153
	(0.097)	(0.195)
High. Inc. $\times u_{i,s}^{HS}$	0.346***	0.186**
	(0.116)	(0.155)
State Controls	No	Yes
Clustered SE state	No	Yes
Obs.	324,353	324,353
R-squared	0.006	0.029
Test Middle ($\beta_1 + \beta_2$)	0.286***	0.357**
Test High ($\beta_1 + \beta_3$)	0.451***	0.390***

Note: Replicates the dropout column of Table 4 using the May unemployment rate as the graduation-year measure. Results are robust. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A18: Transitions from enrolled to dropout and from labor force to enrolled excluding COVID

	College Dropouts	LF → College
$u_{s,t}$	0.105*	-0.055
	(0.061)	(0.079)
Mid. Inc. $\times u_{s,t}$	-0.207***	0.579***
	(0.061)	(0.107)
High. Inc. $\times u_{s,t}$	-0.268***	0.854***
	(0.075)	(0.157)
Obs.	688,976	1,983,714
R-squared	0.004	0.015
Test Middle ($\alpha_1 + \alpha_2$)	-0.102***	0.524***
Test High ($\alpha_1 + \alpha_3$)	-0.163***	0.799***

Note: Replicates Table 2 excluding observations from 2020–2021 to assess sensitivity to the COVID-19 period. The magnitude of the low-income dropout effect falls modestly but remains positive and statistically significant. Effects in percentage points. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A19: College enrollment and completion and the business cycle excluding COVID

	Attended college		College Degree or more	
$u_{i,s}^{HS}$	0.180***	0.271**	-0.080	-0.250**
	(0.060)	(0.119)	(0.063)	(0.114)
Mid. Inc. $\times u_{i,s}^{HS}$	0.527***	0.364	0.643***	0.274
	(0.083)	(0.229)	(0.095)	(0.212)
High. Inc. $\times u_{i,s}^{HS}$	0.819***	0.387	1.781***	0.835***
	(0.087)	(0.265)	(0.115)	(0.200)
State Controls	No	Yes	No	Yes
Clustered SE state	No	Yes	No	Yes
Obs.	647,525	647,525	354,199	354,199
R-squared	0.020	0.058	0.042	0.261
Test Middle ($\beta_1 + \beta_2$)	0.707***	0.635***	0.563***	0.024
Test High ($\beta_1 + \beta_3$)	0.999***	0.658***	1.702***	0.585***

Note: Replicates Table 4 excluding observations from 2020–2021. Results remain similar. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A20: College dropouts and college degree and the business cycle excluding COVID

	College Dropout	
$u_{s,t}$	0.232*** (0.073)	0.325*** (0.105)
Mid. Inc. $\times u_{s,t}$	0.367*** (0.106)	0.297 (0.212)
High. Inc. $u_{s,t}$	0.832*** (0.133)	0.519** (0.241)
State Controls	No	Yes
Clustered SE state	No	Yes
Obs.	316,336	316,336
R-squared	0.006	0.029
Test Middle ($\beta_1 + \beta_2$)	0.599***	0.622***
Test High ($\beta_1 + \beta_3$)	1.064***	0.843***

Note: Replicates the dropout column of Table 4 excluding observations from 2020–2021. Results remain similar. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A21: Transitions from enrolled to dropout and from labor force to enrolled with 23 year old or younger

	College Dropouts	LF $\rightarrow$ College
$u_{s,t}$	0.139** (0.069)	-0.140* (0.082)
Mid. Inc. $\times u_{s,t}$	-0.187*** (0.065)	0.509*** (0.106)
High. Inc. $\times u_{s,t}$	-0.194*** (0.075)	0.655*** (0.140)
Obs.	237,750	1,762,211
R-squared	0.003	0.023
Test Middle ($\alpha_1 + \alpha_2$)	-0.048	0.368***
Test High ($\alpha_1 + \alpha_3$)	-0.055*	0.514***

Note: Replicates Table 4 restricting the sample to individuals aged 23 or younger to mitigate the potential correlation between own earnings and family income. Results remain robust. Effects in percentage points. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A22: College enrollment and completion and the business cycle with 23 year old or younger

	Attended college		College Degree or more	
$u_{i,s}^{HS}$	0.217***	0.330**	-0.175***	-0.185
	(0.066)	(0.129)	(0.064)	(0.124)
Mid. Inc. $\times u_{i,s}^{HS}$	0.462***	0.378*	0.341***	0.227
	(0.094)	(0.195)	(0.096)	(0.161)
High. Inc. $\times u_{i,s}^{HS}$	0.690***	0.323	1.338***	0.735***
	(0.097)	(0.238)	(0.125)	(0.183)
State Controls	No	Yes	No	Yes
Clustered SE state	No	Yes	No	Yes
Obs.	532,803	532,803	275,928	275,928
R-squared	0.018	0.062	0.026	0.242
Test Middle ($\beta_1 + \beta_2$)	0.678***	0.708***	0.166**	0.042
Test High ($\beta_1 + \beta_3$)	0.907***	0.653***	1.163***	0.550***

Note: Replicates Table 4 restricting the sample to individuals aged 23 or younger. Results remain robust, confirming that family income is approximately uncorrelated with own earnings at these ages. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A23: College dropouts and college degree and the business cycle with 23 year old or younger

	College Dropout	
$u_{s,t}$	0.231***	0.378***
	(0.080)	(0.127)
Mid. Inc. $\times u_{s,t}$	0.267**	0.215
	(0.116)	(0.201)
High. Inc. $u_{s,t}$	0.723***	0.399*
	(0.146)	(0.210)
State Controls	No	Yes
Clustered SE state	No	Yes
Obs.	262,485	262,485
R-squared	0.007	0.036
Test Middle ($\beta_1 + \beta_2$)	0.498***	0.591***
Test High ($\beta_1 + \beta_3$)	0.954***	0.774***

Note: Replicates the dropout column of Table 4 restricting the sample to individuals aged 23 or younger. Results remain robust. Effects in percentage points relative to having only a high school diploma. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A24: Transitions controlling for state tuition and financial aid

	College Dropouts	LF $\rightarrow$ College
$u_{s,t}$	0.215*** (0.073)	-0.215** (0.100)
Mid. Inc. $\times u_{s,t}$	-0.195*** (0.071)	0.502*** (0.118)
High. Inc. $\times u_{s,t}$	-0.175** (0.080)	0.560*** (0.156)
Log avg. public 4-yr tuition $_{s,t}$	-2.793*** (0.560)	4.117*** (0.814)
Log state need-based grant aid per FTE $_{s,t}$	0.252 (0.174)	-0.065 (0.179)
Obs.	438,367	376,335
R-squared	0.004	0.014
Controls	Yes	Yes
Test Middle ($\alpha_1 + \alpha_2$)	0.020	0.287***
Test High ($\alpha_1 + \alpha_3$)	0.040	0.345***

Note: Replicates the preferred specification of Table 2 adding log average published in-state tuition at public four-year institutions and log state need-based grant aid per undergraduate FTE as additional controls. Average tuition data are from the College Board Trends in College Pricing series, available from 2004 onward, which restricts the sample to 2004–2024 and accounts for the reduction in observations relative to Table 2. Tuition prices are reported by academic year and merged to the CPS monthly data by state, with the 2004–2005 academic year price assigned to observations from September 2004 through August 2005 and so on. State need-based grant aid per undergraduate FTE is from the NASSGAP Annual Survey of State Student Financial Aid, merged by state and year. Effects in percentage points relative to staying in college or staying in the labor force. Panel regression using random effects. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A25: College enrollment and completion controlling for state tuition and financial aid

	Attended college (1)	College Degree or more (2)	College Dropout (3)
$u_{i,s}^{HS}$	0.067 (0.108)	-0.251** (0.096)	0.222** (0.108)
Mid. Inc. $\times u_{i,s}^{HS}$	0.246 (0.185)	0.040 (0.195)	0.088 (0.182)
High. Inc. $\times u_{i,s}^{HS}$	0.291 (0.205)	0.686*** (0.155)	0.308* (0.176)
Log avg. public 4-yr tuition $_{s,t}$	9.711*** (1.218)	3.220** (0.385)	10.584*** (1.502)
Log state need-based grant aid per FTE $_{s,t}$	0.224 (0.308)	0.744* (0.385)	-0.569* (0.306)
State Controls	Yes	Yes	Yes
Clustered SE state	Yes	Yes	Yes
Obs.	402,337	223,651	185,422
R-squared	0.062	0.267	0.032
Test Middle ($\beta_1 + \beta_2$)	0.313**	-0.211	0.310
Test High ($\beta_1 + \beta_3$)	0.359**	0.435**	0.530***

Note: Replicates Table 4 adding log average published in-state tuition at public four-year institutions and log state need-based grant aid per undergraduate FTE as additional controls, restricting the sample to individuals who graduated high school from 2004 onward due to data availability. Tuition is from the College Board Trends in College Pricing series and assigned by academic year of high school graduation; aid is from the NASSGAP Annual Survey, merged by state and graduation year. All three columns include state fixed effects and standard errors clustered at the state level. Effects in percentage points relative to having only a high school diploma. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A26: Transitions with state-specific linear time trends

	College Dropouts	LF → College
$u_{s,t}$	0.151** (0.064)	-0.087 (0.081)
Mid. Inc. $\times u_{s,t}$	-0.197*** (0.062)	0.505*** (0.106)
High. Inc. $\times u_{s,t}$	-0.207*** (0.070)	0.617*** (0.106)
Obs.	706,702	2,024,651
R-squared	0.004	0.015
Controls	Yes	Yes
Test Middle ($\alpha_1 + \alpha_2$)	-0.045	0.418***
Test High ($\alpha_1 + \alpha_3$)	-0.056*	0.530***

Note: Replicates Table 2 adding a state-specific linear time trend for each state, which absorbs any unobserved gradual policy changes or secular trends at the state level that may be correlated with state unemployment rates. The full sample is retained. Results are virtually identical to the baseline. Effects in percentage points relative to staying in college or staying in the labor force. Panel regression using random effects. Standard errors clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A27: College enrollment and completion with state-specific linear time trends

	Attended college (1)	College Degree or more (2)	College Dropout (3)
$u_{i,s}^{HS}$	0.151 (0.132)	-0.309** (0.126)	0.235** (0.117)
Mid. Inc. $\times u_{i,s}^{HS}$	0.358 (0.242)	0.275 (0.238)	0.267 (0.229)
High. Inc. $\times u_{i,s}^{HS}$	0.368 (0.267)	0.945*** (0.210)	0.414* (0.226)
State Controls	Yes	Yes	Yes
Clustered SE state	Yes	Yes	Yes
Obs.	665,552	364,787	324,353
R-squared	0.059	0.263	0.030
Test Middle ($\beta_1 + \beta_2$)	0.509***	-0.034	0.503**
Test High ($\beta_1 + \beta_3$)	0.519***	0.636***	0.650***

Note: Replicates Table 4 adding a state-specific linear time trend for each state, which absorbs any unobserved gradual policy changes or secular trends at the state level correlated with state unemployment rates around high school graduation. The full sample is retained. All three columns include state fixed effects and standard errors clustered at the state level. Effects in percentage points relative to having only a high school diploma. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A28: Summary of model results with $\theta(D) = 1.013 > \theta(N) = 1$

Income level	Expansion ($z = z_{high}$)			Recession ($z = z_{low}$)		
	Low	Middle	High	Low	Middle	High
Result 1: Financial Dropouts						
% of E^+ who drop out	0	0	0	14.3	0	0
Δ w.r.t expansion				+14.3	0	0
Result 2: Cyclicity of immediate college enrollment						
% enrolling at $\tau = 0$ enrollment	54.3	89	70.4	35.4	87.5	87.2
Δ w.r.t expansion				-18.9	-1.5	+16.8
Result 3: Immediate enrollment probability p threshold						
Min p to enroll at $\tau = 0$	47	11.5	15.2	65.6	13	10.8
Δ w.r.t expansion				+18.6	+1.5	-4.4
Result 4: Strategic delays						
% enrolling at $\tau = 1$ enrollment	0	0	15.2	0	0	2.4
Δ w.r.t expansion				0	0	-12.8

Note: Replicates Table 6 setting $\theta(D) = 1.013 > \theta(N) = 1$. The value is calibrated by applying to college dropouts the same ratio of raw to causal earnings premium used for college graduates: Heckman et al. (2018) estimate a causal premium of 20% against a raw premium of 62%, giving a scaling factor of 3.1; applying this to the 4% raw dropout premium yields $\theta(D) = 1.013$. Main qualitative results are unchanged.

Table A29: Summary of results of the model with $\theta(C)^H > \theta(C)^M > \theta(C)^L$

Income level	Expansion ($z = z_{high}$)			Recession ($z = z_{low}$)		
	Low	Middle	High	Low	Middle	High
Result 1: Financial Dropouts						
% of E^+ who drop out	0	0	0	14.3	0	0
Δ w.r.t expansion				+14.3	0	0
Result 2: Cyclicalilty of immediate college enrollment						
% enrolling at $\tau = 0$ enrollment	52	86.5	72	37.7	86	88
Δ w.r.t expansion				-14.3	-0.5	+16
Result 3: Immediate enrollment probability p threshold						
Min p to enroll at $\tau = 0$	55.3	14.1	15	63.7	14.6	11.7
Δ w.r.t expansion				+8.4	+0.5	-3.3
Result 4: Strategic delays						
% enrolling at $\tau = 1$ enrollment	0	0	13.6	0	0	0.8
Δ w.r.t expansion				0	0	-12.8

Note: Replicates Table 6 allowing income-group-specific college wage premiums: $\theta(C)^L = 1.15$, $\theta(C)^M = 1.20$, $\theta(C)^H = 1.25$. The tuition parameter is recalibrated to $f = 0.787$ (full tuition = 116% of $\bar{y}^{NC}$) to maintain the enrollment disparity moment from Table 3. Main qualitative results are unchanged.

Table A30: Policy relevance: The role of p , f and ψ in terms of present value of consumption

Income level	% Loss in Expected Lifetime Present Value Consumption in a Recession					
	While enrolled			Around HS graduation		
	Low	Middle	High	Low	Middle	High
1) <u>Prob. of good fit p</u>						
$p = 12.5\%$	-	-	-	7.99	6.06	5.01
Baseline averaging across p	-	-	-	8.49	5.72	4.42
$p = 100\%$	-	-	-	9.71	5.46	3.21
2) <u>Tuition costs f</u>						
$f = 100\%\bar{y}^{NC}$	8.89	5.52	4.63	8.60	5.65	4.26
Baseline $f = 160\%\bar{y}^{NC}$	8.82	5.57	4.67	8.49	5.72	4.42
$f = 200\%\bar{y}^{NC}$	10.31	5.61	4.69	8.34	5.78	4.51
3) <u>Labor market entry conditions ψ</u>						
$\psi = 0$	8.88	5.64	4.72	8.40	5.68	4.38
Baseline $\psi = 0.12$	8.82	5.57	4.67	8.49	5.72	4.42
$\psi = 0.5$	8.65	5.37	4.50	8.82	5.95	4.60

Note: Replicates Table 7 expressing losses as percentage reductions in expected lifetime present value consumption rather than lifetime utility. The gap between income groups is narrower than in the utility-based table, reflecting the concavity of log utility which amplifies losses at low consumption levels. All other definitions follow Table 7.

B Mathematical Appendix

B.1 Derivation of p and θ thresholds

To derive the threshold values for p and θ at which an individual would be indifferent between going to college and entering the labor force directly, we need to set the expected incomes Y^C and Y^N equal to each other:

$$Y^C = Y^N$$

Given the equations:

$$\begin{aligned} Y^C &= -f + p \left(-f + \theta(C) \mathbb{E}_0 \sum_{\tau=2}^{\mathcal{T}} [\psi_{\tau} \exp(z_2) + (1 - \psi_{\tau}) \exp(z_{\tau})] \right) \\ &\quad + (1 - p) \left(\theta(D) \mathbb{E}_0 \sum_{\tau=1}^{\mathcal{T}} [\psi_{\tau} \exp(z_1) + (1 - \psi_{\tau}) \exp(z_{\tau})] \right) \\ Y^N &= \theta(N) \mathbb{E}_0 \sum_{\tau=0}^{\mathcal{T}} [\psi_{\tau} \exp(z_0) + (1 - \psi_{\tau}) \exp(z_{\tau})] \end{aligned}$$

Set $Y^C = Y^N$:

$$\begin{aligned} &-f + p \left(-f + \theta(C) \mathbb{E}_0 \sum_{\tau=2}^{\mathcal{T}} [\psi_{\tau} \exp(z_2) + (1 - \psi_{\tau}) \exp(z_{\tau})] \right) \\ &\quad + (1 - p) \left(\theta(D) \mathbb{E}_0 \sum_{\tau=1}^{\mathcal{T}} [\psi_{\tau} \exp(z_1) + (1 - \psi_{\tau}) \exp(z_{\tau})] \right) \\ &= \theta(N) \mathbb{E}_0 \sum_{\tau=0}^{\mathcal{T}} [\psi_{\tau} \exp(z_0) + (1 - \psi_{\tau}) \exp(z_{\tau})] \end{aligned}$$

To simplify, let's denote:

$$\begin{aligned} \Lambda^2 &= \mathbb{E}_0 \sum_{\tau=2}^{\mathcal{T}} [\psi_{\tau} \exp(z_2) + (1 - \psi_{\tau}) \exp(z_{\tau})] \\ \Lambda^1 &= \mathbb{E}_0 \sum_{\tau=1}^{\mathcal{T}} [\psi_{\tau} \exp(z_1) + (1 - \psi_{\tau}) \exp(z_{\tau})] \\ \Lambda^0 &= \mathbb{E}_0 \sum_{\tau=0}^{\mathcal{T}} [\psi_{\tau} \exp(z_0) + (1 - \psi_{\tau}) \exp(z_{\tau})] \end{aligned}$$

The equation becomes:

$$-f + p(-f + \theta(C)\Lambda^2) + (1-p)(\theta(D)\Lambda^1) = \theta(N)\Lambda^0$$

Expanding and defining $\hat{p}$ as the p such that $Y^C = Y^N$:

$$-f - \hat{p}f + \hat{p}\theta(C)\Lambda^2 + \theta(D)\Lambda^1 - \hat{p}\theta(D)\Lambda^1 = \theta(N)\Lambda^0$$

Rearranging:

$$-\hat{p}f + \hat{p}\theta(C)\Lambda^2 - \hat{p}\theta(D)\Lambda^1 = \theta(N)\Lambda^0 + f - \theta(D)\Lambda^1$$

$$\hat{p}(\theta(C)\Lambda^2 - \theta(D)\Lambda^1 - f) = \theta(N)\Lambda^0 - \theta(D)\Lambda^1 + f$$

Solving for $\hat{p}$:

$$\hat{p} = \frac{\theta(N)\Lambda^0 - \theta(D)\Lambda^1 + f}{\theta(C)\Lambda^2 - \theta(D)\Lambda^1 - f}$$

This equation gives the threshold value for p at which an individual would have the same lifetime income if she decides to go to college and if she decides to enter the labor force directly.

I do the same for the derivation of $\hat{\theta}(C)$ which is college wage premium such that $Y^C = Y^N$.

Starting from:

$$-pf + p\hat{\theta}(C)\Lambda^2 - p\theta(D)\Lambda^1 = \theta(N)\Lambda^0 + f - \theta(D)\Lambda^1$$

Rearranging:

$$p\hat{\theta}(C)\Lambda^2 = \theta(N)\Lambda^0 - \theta(D)\Lambda^1 + p\theta(D)\Lambda^1 + f + pf$$

$$p\hat{\theta}(C)\Lambda^2 = \theta(N)\Lambda^0 - (1-p)\theta(D)\Lambda^1 + (1+p)f$$

Solving for $\hat{\theta}(C)$:

$$\hat{\theta}(C) = \frac{\theta(N)\Lambda^0 - (1-p)\theta(D)\Lambda^1 + (1+p)f}{p\Lambda^2}$$